

An exact relaxation method for complementarity constraints of energy storages in power grid optimization problems

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HIGHLIGHTS

- An exact generalized relaxation condition (GRC) with a priori property is proposed for complementarity constraints of energy storages.
- The proposed GRC has a broader scope than the existing relaxation conditions and is applicable to various power system optimization problems.
- By introducing penalties for charging and discharging losses of storage into the objective, a practical relaxation method is developed based on the proposed GRC.
- The applicability of the proposed GRC can be ensured for scenarios with arbitrary electricity prices by adjusting the penalty parameter.

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ABSTRACT

The increasing deployment of energy storages in power grid necessitates the consideration of their operational costs and constraints. However, energy storages introduce complementary constraints or binary variables, make the optimization problems non-convex and challenging to solve. To tackle this issue, we propose a generalized relaxation condition (GRC), which possesses a priori property. As long as the GRC is met, it is guaranteed that there is no simultaneous charging and discharging (SCD) in the relaxed optimization problems excluding the complementarity constraints. Moreover, we prove that the exact relaxation region on the locational marginal prices (LMPs) formed by the proposed GRC contains those of the other existing relaxation conditions. The proposed GRC can be satisfied for the majority of electricity price scenarios, including positive prices and mild negative prices. Beyond that, through introducing penalties for charging and discharging losses of storage into the objective function, the proposed GRC can be further ensured in all price scenarios. We also develop a criterion to set the penalty parameter properly, making the GRC satisfied. Afterward, a practical relaxation method is presented based on the proposed GRC. Finally, we propose a practical two-stage penalty parameter adjustment scheme to weaken the dependence of the proposed relaxation method on the prediction accuracy of the LMPs. The proposed exact relaxation condition is applicable to various scenarios, such as storage-participating economic dispatch, joint energy-reserve optimization and Volt/VAR optimization. Finally, comprehensive test cases verify the exactness and advantages of the proposed relaxation method.

1. Introduction

1.1. Background

Recently, there has been a rising trend in storage-plus-generation co-located hybrid resources, such as renewable energy paired with

storages, and thermal units equipped with storages [1]. Furthermore, the Federal Energy Regulatory Commission (FERC) of USA issued Order 841 in 2018 [2], which requires that each Regional Transmission Organization (RTO) and Independent System Operator (ISO) develops a wholesale market participation model to facilitate the participation of storages in energy and ancillary services, through including storages' parameters that represent their distinct physical and operational

Abbreviations: GRC, Generalized relaxation condition; SCD, Simultaneous charging and discharging; LMP, Locational marginal price; ESS, Energy storage system; MIP, Mixed-integer programming; ED, Economic dispatch; RG, Renewable generation; SOC, State of Charge.

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Nomenclature

Sets

- $\mathcal{N}, \mathcal{E}$ The bus set and branch set
 $\mathcal{T}$ Index set of the whole optimization horizon
 $\mathcal{S}_{\text{ESS}}$ Index set of storage

Parameters

- T The whole optimization horizon
 Δt Optimization time interval
 $\bar{P}_{\text{ESS},n}^{\text{ch}}, \bar{P}_{\text{ESS},n}^{\text{dc}}$ The upper limits of charging and discharging active power of the n^{th} ESS
 $E_{\text{ESS},n,0}$ The initial SOC level of the n^{th} ESS
 $\delta_{\text{ESS},n}$ The self-discharge rate of the n^{th} ESS
 $\eta_{\text{ESS},n}^{\text{ch}}, \eta_{\text{ESS},n}^{\text{dc}}$ The charging and discharging efficiency of the n^{th} ESS
 $\underline{E}_{\text{ESS},n}, \bar{E}_{\text{ESS},n}$ Lower and upper bounds of the SOC level of the n^{th} ESS

Variables

- $P_{ij,t}, Q_{ij,t}$ Active and reactive power flow of line ij at time t
 $P_{j,t}, Q_{j,t}$ Net injected active and reactive power at bus j and time t
 $P_{\text{ESS},n,t}^{\text{dc}}, P_{\text{ESS},n,t}^{\text{ch}}$ Discharging and charging active power of the n^{th} ESS at time t
 $q_{\text{ESS},n,t}, S_{\text{ESS},n}$ The reactive power and apparent power of the n^{th} ESS
 $R_{n,t}^{\text{ch,ESS-}}, R_{n,t}^{\text{ch,ESS+}}$ Downward/upward reserves of the n^{th} storage when it is in the charging status at time t
 $R_{n,t}^{\text{dc,ESS-}}, R_{n,t}^{\text{dc,ESS+}}$ Downward/upward reserves of the n^{th} storage when it is in the discharging status at time t
 $E_{\text{ESS},n,t}$ The SOC level of the n^{th} ESS at time t
 $z_{\text{ESS},n,t}^{\text{ch}}, z_{\text{ESS},n,t}^{\text{dc}}$ The binary variables indicating the charging and discharging state of the n^{th} ESS at time t

characteristics. In summary, energy storage is playing an increasing important role in the operation of power systems.

In the conventional sense, the complementarity constraints of energy storage systems (ESSs) are introduced to avoid simultaneous charging and discharging (SCD), which render the whole optimization problem non-convex and challenging to solve. To cope with this challenge, binary variables are usually employed to transform the complementarity constraints into mixed-integer programming (MIP). For instance, NYISO introduced its storage modeling within the dispatch model, incorporating binary variables for battery storages [3]. Besides, there is already a considerable body of literature exploring the involvement of energy storage in the joint energy-reserve optimization, most of which introduced integer variables to avoid SCD [4–7]. However, the MIP remains computationally complex as the number of ESSs or the time periods increases in multi-period optimization problems [8].

1.2. Previous research

Recently, some literature has discussed the relaxation techniques for the complementarity constraints of ESSs, which can be roughly classified into four categories:

i) In the first category, the complementarity constraints are reformulated as linear constraints about the charging/discharging active power and their upper bounds [9–11]. Nevertheless, the formulations still allow a significant degree of simultaneous charging and discharging, that is, they cannot theoretically guarantee that SCD will be avoided.

ii) Feasible solution recovery strategies were introduced in [12,13], with which the complementarity constraints could be met. Besides, the authors in [14] supposed that charging and discharging efficiencies were both 1 to acquire a solution, which was probably not feasible in practice, and then readjusted the power of electric vehicles (EVs) for a new feasible solution. However, these methods can only ensure a sub-optimal feasible solution, along with a double computational burden.

iii) Based upon certain assumptions on the charging and discharging fees, some other relaxation methods were developed in [15–18]. Specifically, [16] assumed an increasing positive operational cost function for both charging and discharging. However, this assumption does not align with the energy storage arbitrage scenario, wherein the storage owner paying the grid for charging energy while receiving compensation from the grid for discharging. [15,18] provided analyses that more generation is needed to deliver the same power when charging and discharging simultaneously, combining with the fact that both charging and discharging efficiencies are less than 1. Correspondingly, the generation cost is also higher when SCD occurs. A method to avoid the SCD by modifying the objective function was proposed in [19], and some qualitative analyses were provided. However, in scenarios with high renewable energy generation, all the qualitative conclusions drawn from these analyses face growing limitations in their applicability.

iv) Some studies employ KKT conditions to analyze under which conditions the optimal solution satisfies the complementarity constraints [8,20–24]. Wherein sufficient relaxation conditions were proposed for the economic dispatch (ED) problem of transmission networks in [20,21]. And an analysis was provided in [22] on the relaxation of grid-connected ESSs in a DCOPF problem for model predictive control. While the conditions presented in [22] not only rely on some strong assumptions on the objective function, but also are a posteriori, which limits their applications. As for ACOPF, the interaction between LMPs and the optimal operation of ESSs was studied in [23]. Besides, [8,24] provided preliminary analyses for the distribution network (DN) loss-minimizing problems, and derived sufficient conditions. While the condition C3 in [8] is rather restrictive, especially under the high penetration of renewable energy [8].

It's also noteworthy that the majority of existing relaxation conditions all require that the LMPs of the storage-connected buses should be non-negative, such as, the *relaxation condition of group C* in [21], and the *relaxation conditions* in [22–24]. Even though these conditions can be met in most of the cases, a growing number of negative electricity price scenarios appear with increasing renewable energy integration and transmission congestion. For example, [25] shows that the share of negative electricity prices is nearly 10% in February 2022 in the Real-Time Market of ERCOT, as shown in Table 1. In such cases, the aforementioned relaxation conditions may no longer apply. Only a few studies [20,21] provided relaxation conditions that enable the existence of slightly negative LMPs. However, with the large-scale integration of renewable energy, the occurrences of negative LMPs will become more frequent and severe, making all of these relaxation conditions

Table 1
Percentages of Negative LMPs in the RTM of ERCOT in 2022 by Month.

Month	LMP < 0 \$/MWh	LMP < -10 \$/MWh	LMP < -20 \$/MWh	LMP < -30 \$/MWh
1	2.99%	0.34%	0.06%	0.002%
2	<u>9.62%</u>	<u>3.64%</u>	<u>1.79%</u>	<u>0.22%</u>
3	6.74%	1.66%	0.54%	0.01%
4	4.84%	1.67%	0.69%	0.006%
5	4.25%	0.34%	0.06%	0.009%
6	1.52%	0.005%	0.00%	0.00%
7	0.004%	0.00%	0.00%	0.00%
8	0.006%	0.00%	0.00%	0.00%
9	0.42%	0.29%	0.17%	0.008%
10	2.74%	0.58%	0.18%	0.00%
11	4.30%	0.80%	0.14%	0.038%
12	6.42%	0.75%	0.20%	0.00%

increasingly difficult to apply.

Considering that renewable energy and transmission congestion are the major incentives for the deployment of storages, the negative LMP scenarios can be expected to occur occasionally. In this context, there are still some limitations in existing research. We provide a summary by method category, as shown in Table 2. In summary, a more general exact relaxation method for power grid optimization problems should be investigated. The relaxation method should have the following characteristics:

- 1) It should possess the capability to ensure the exact convex relaxation of storage models under a variety of negative price scenarios;
- 2) It should have a prior property to facilitate practical application;
- 3) It should not depend on the specific form of the objective function and the operational constraints within the optimization problem, thus ensuring the general applicability of the proposed relaxation method across various application scenarios, such as storage-participating ED, joint energy-reserve optimization and Volt/VAR optimization.

1.3. Contributions

The contributions of this work are summarized as follows:

- 1) An exact generalized relaxation condition (GRC) with a priori property is firstly proposed that provably ensure no SCD in the relaxed optimization problems excluding the complementarity constraints. Besides, the proposed GRC holds for various application scenarios, such as storage-participating ED, joint energy-reserve optimization and Volt/VAR optimization. Moreover, the proposed GRC can be satisfied for the majority of electricity price scenarios, including positive prices and mild negative prices. We then prove that the exact relaxation region about LMPs formulated by the proposed GRC contains those of the other existing ones in [20–24]. To the best of our knowledge, we are the first to establish the relative inclusion relationships among multiple energy storage relaxation conditions.
- 2) By introducing penalties for charging and discharging losses of storage into the objective function, the applicability of the proposed GRC can be extended to the scenarios with arbitrary electricity prices. Besides, we propose a criterion for determining the penalty parameter, ensuring the satisfaction of the GRC.

Table 2
Summarizing the Limitations of Existing Methods by Categories.

Relaxation methods	Limitations
Category i)	This category of method narrows the feasible region of charging and discharging power by introducing additional constraints. While these formulations still allow a significant degree of SCD, that is, they cannot theoretically guarantee that SCD will be avoided.
Category ii)	Methods that fall into this category first provide a lower bound on the true optimal objective value. Then, the recovery strategy is employed to generate a feasible solution. However, such methods can only ensure a suboptimal feasible solution, along with a double computational burden.
Category iii)	Methods of this kind depend on certain assumptions about the charging and discharging rates. In scenarios with high level of renewable energy generation, all the qualitative conclusions drawn from these analyses are increasingly limited in their applicability.
Category iv)	In this category, the majority of existing relaxation conditions require that the LMPs of the storage-connected buses should be non-negative. Only a few studies have provided relaxation conditions that allow slightly negative LMPs. However, with the large-scale integration of renewable energy, the occurrences of negative LMPs will become more frequent and severe, making all of these relaxation conditions increasingly difficult to apply.

- 3) Given a priori characteristic of the proposed GRC, a practical relaxation method is presented for power grid optimization problems considering energy storages. In occasional scenarios where the forecasted LMPs substantially differ from their real values, we further propose a practical two-stage penalty parameter adjustment scheme. This involves increasing the penalty parameter to guarantee compliance with the GRC, thereby enhancing the exactness of the relaxation. Through integrating the aforementioned scheme, the dependency of the proposed relaxation method on the prediction accuracy of the LMPs has been significantly reduced.

The rest of this paper is structured as follows: Section II describes a compact model for the power grid optimization problems. In Section III, we derive the GRC, followed by the inclusion relationships between the proposed GRC and the other existing ones. In Section IV, the applicability of the proposed GRC is further broadened through introducing penalties for charging and discharging losses of storage into the objective function. Besides, we establish a criterion for appropriately determining the penalty parameter. Afterward, a practical relaxation method is presented based on the proposed GRC. Case studies are conducted in Section V. Section VI summarizes the conclusions.

2. Mathematical model formulation

This section introduces a compact model for the storage-participating power grid optimization problems. Considering that the central theme of this work is to propose the exact relaxation method for complementarity constraints of energy storages in power grid optimization problems, thus problem (P1) should incorporate specific operational constraints relevant to energy storage systems, as demonstrated in the later part of this section.

$$(P1) \min_{\mathbf{p}^{\text{ch}}, \mathbf{p}^{\text{dc}}, \mathbf{x}} \text{obj} = F(\mathbf{p}^{\text{ch}}, \mathbf{p}^{\text{dc}}, \mathbf{x}) \quad (1)$$

$$\begin{aligned} s.t. & G(\mathbf{p}^{\text{ch}}, \mathbf{p}^{\text{dc}}, \mathbf{x}) \leq 0 \\ & 0 \leq \mathbf{p}^{\text{ch}} \leq \bar{\mathbf{p}}^{\text{ch}} \\ & 0 \leq \mathbf{p}^{\text{dc}} \leq \bar{\mathbf{p}}^{\text{dc}} \\ & \underline{\mathbf{x}} \leq \mathbf{x} \leq \bar{\mathbf{x}} \end{aligned} \quad (2)$$

$$\mathbf{p}^{\text{ch}} \circ \mathbf{p}^{\text{dc}} = \mathbf{0} \quad (3)$$

where $\mathbf{p}^{\text{ch}} = \{p_{\text{ESS},n,t}^{\text{ch}}, \forall n, \forall t\}$ and $\mathbf{p}^{\text{dc}} = \{p_{\text{ESS},n,t}^{\text{dc}}, \forall n, \forall t\}$ are charging and discharging power of storages; $\mathbf{x}$ is the vector composed of other decision variables; $F(\mathbf{p}^{\text{ch}}, \mathbf{p}^{\text{dc}}, \mathbf{x})$ is the continuously differentiable objective function; $G(\mathbf{p}^{\text{ch}}, \mathbf{p}^{\text{dc}}, \mathbf{x})$ represent the sets of inequality constraints; $\mathbf{0}$ is a vector whose elements are all zero; $\bar{\mathbf{p}}^{\text{ch}}$ and $\bar{\mathbf{p}}^{\text{dc}}$ are upper limits of charging and discharging power; $\underline{\mathbf{x}}$ and $\bar{\mathbf{x}}$ are lower and upper limits of $\mathbf{x}$; constraint (3) represents the complementarity constraint of storages; the operation $\circ$ denotes the Hadamard product.

Assumptions: Three specific assumptions are given for (P1):

- i) The (local) optimal solution for problem (P1) satisfies the KKT conditions.
- ii) The optimization problem (P1) satisfies the constraint qualifications.
- iii) The terms in the objective function (1) regarding the charging and discharging active power of ESSs are continuously differentiable.

The derivations and analyses presented in this work are based on the KKT conditions, necessitating the fulfillment of *Assumption i)*, which is also a common assumption in existing literature [20–24]. The KKT conditions are necessary conditions for the (local) optimal solution of optimization problem (P1) only if the constraint qualifications are satisfied. Thus, *Assumption ii)* is indispensable. The *Assumption iii)* is also

easy to keep. For instance, most literature presents energy storage arbitrage in a linear form [10,16,20,21].

We then provide a detailed formulations of the nodal power balance constraint related to LMP and these operational constraints of ESSs, to facilitate the subsequent descriptions and proof of the proposed GRC. For simplicity, other constraints are not explicitly presented here.

1) Nodal power balance constraints

$$p_{j,t} - \sum_{i:j \in \mathcal{E}} p_{ji,t} - \sum_{i:j \in \mathcal{E}} p_{ji,t} = 0, \forall j \in \mathcal{N}, \forall t \quad (\lambda_{j,t}^p) \quad (4)$$

$$q_{j,t} - \sum_{i:j \in \mathcal{E}} q_{ji,t} - \sum_{i:j \in \mathcal{E}} q_{ji,t} = 0, \forall j \in \mathcal{N}, \forall t \quad (5)$$

where $\lambda_{j,t}^p$ is the multiplier of the active power balance constraint at bus j and time t .

2) Operational constraints of ESSs

To take full advantage of the four-quadrant operation capabilities of ESSs, their operation should comply with the following constraints.

$$-p_{ESS,n,t}^{\text{ch}} \leq 0, p_{ESS,n,t}^{\text{ch}} \leq \bar{P}_{ESS,n}^{\text{ch}}, \quad (\lambda_{ESS,n,t}^{\text{ch},1}, \lambda_{ESS,n,t}^{\text{ch},2}) \quad (6)$$

$$-p_{ESS,n,t}^{\text{dc}} \leq 0, p_{ESS,n,t}^{\text{dc}} \leq \bar{P}_{ESS,n}^{\text{dc}}, \quad (\lambda_{ESS,n,t}^{\text{dc},1}, \lambda_{ESS,n,t}^{\text{dc},2}) \quad (7)$$

$$(p_{ESS,n,t}^{\text{dc}})^2 + (q_{ESS,n,t})^2 \leq (S_{ESS,n})^2, \quad (\lambda_{ESS,n,t}^{S,1}) \quad (8)$$

$$(p_{ESS,n,t}^{\text{ch}})^2 + (q_{ESS,n,t})^2 \leq (S_{ESS,n})^2, \quad (\lambda_{ESS,n,t}^{S,2}) \quad (9)$$

where $\lambda_{ESS,n,t}^{\text{ch},1}, \lambda_{ESS,n,t}^{\text{ch},2}, \lambda_{ESS,n,t}^{\text{dc},1}, \lambda_{ESS,n,t}^{\text{dc},2}, \lambda_{ESS,n,t}^{S,1}$ and $\lambda_{ESS,n,t}^{S,2}$ are the multipliers of the corresponding constraints.

Moreover, ESSs can provide upward reserves by either reducing the charging power or increasing the discharging power. Similarly, downward reserve capacity can be provided by charging more or discharging less from their predetermined power schedules, as shown in Fig. 1 [30]. Correspondingly, constraints (10)–(14) together ensure the achievability of reserves from storages by considering both the charging/discharging power and energy limits [20,30].

$$p_{ESS,n,t}^{\text{ch}} + R_{n,t}^{\text{ch},\text{ESS}-} \leq \bar{P}_{ESS,n}^{\text{ch}}, \quad (\lambda_{ESS,n,t}^{\text{RD},1}) \quad (10)$$

$$R_{n,t}^{\text{dc},\text{ESS}-} \leq p_{ESS,n,t}^{\text{dc}}, \quad (\lambda_{ESS,n,t}^{\text{RD},2})$$

$$p_{ESS,n,t}^{\text{dc}} + R_{n,t}^{\text{dc},\text{ESS}+} \leq \bar{P}_{ESS,n}^{\text{dc}}, \quad (\lambda_{ESS,n,t}^{\text{RU},1})$$

$$R_{n,t}^{\text{ch},\text{ESS}+} \leq p_{ESS,n,t}^{\text{ch}}, \quad (\lambda_{ESS,n,t}^{\text{RU},2}) \quad (11)$$

$$R_{n,t}^{\text{ch},\text{ESS}-}, R_{n,t}^{\text{dc},\text{ESS}-}, R_{n,t}^{\text{ch},\text{ESS}+}, R_{n,t}^{\text{dc},\text{ESS}+} \geq 0 \quad (12)$$

$$(1 - \delta_{ESS,n})^t E_{ESS,n,0} + \sum_{\tau=1}^t (1 - \delta_{ESS,n})^{t-\tau} \times$$

$$\left((p_{ESS,n,\tau}^{\text{ch}} - R_{n,\tau}^{\text{ch},\text{ESS}+}) \cdot \eta_{ESS,n}^{\text{ch}} - (p_{ESS,n,\tau}^{\text{dc}} + R_{n,\tau}^{\text{dc},\text{ESS}+}) / \eta_{ESS,n}^{\text{dc}} \right) \Delta t \geq E_{ESS,n}, \quad (\lambda_{ESS,n,t}^{\text{SOC},1}) \quad (13)$$

$$(1 - \delta_{ESS,n})^t E_{ESS,n,0} + \sum_{\tau=1}^t (1 - \delta_{ESS,n})^{t-\tau} \times$$

$$\left((p_{ESS,n,\tau}^{\text{ch}} + R_{n,\tau}^{\text{ch},\text{ESS}-}) \cdot \eta_{ESS,n}^{\text{ch}} - (p_{ESS,n,\tau}^{\text{dc}} - R_{n,\tau}^{\text{dc},\text{ESS}-}) / \eta_{ESS,n}^{\text{dc}} \right) \Delta t \leq E_{ESS,n}, \quad (\lambda_{ESS,n,t}^{\text{SOC},2}) \quad (14)$$

where $\lambda_{ESS,n,t}^{\text{RD},1}, \lambda_{ESS,n,t}^{\text{RD},2}, \lambda_{ESS,n,t}^{\text{RU},1}, \lambda_{ESS,n,t}^{\text{RU},2}, \lambda_{ESS,n,t}^{\text{SOC},1}, \lambda_{ESS,n,t}^{\text{SOC},2}$ are the multipliers of the corresponding constraints.

The ESSs' SOC at the initial and final scheduling intervals are restricted to be consistent to guarantee a reliable energy planning for the next T-period dispatch cycle, as shown in (15).

$$E_{ESS,n,T} = (1 - \delta_{ESS,n})^T E_{ESS,n,0} + \sum_{\tau=1}^T (1 - \delta_{ESS,n})^{T-\tau} \times$$

$$\left((p_{ESS,n,\tau}^{\text{ch}} - R_{n,\tau}^{\text{ch},\text{ESS}+}) \cdot \eta_{ESS,n}^{\text{ch}} - (p_{ESS,n,\tau}^{\text{dc}} + R_{n,\tau}^{\text{dc},\text{ESS}+}) / \eta_{ESS,n}^{\text{dc}} \right) \Delta t = E_{ESS,n,0} \quad (15)$$

where $E_{ESS,n,T}$ is the SOC corresponding to time period T .

Besides, we adopt (16) as an enhanced complement to the aforementioned ESS model [9].

$$\frac{p_{ESS,n,t}^{\text{ch}}}{\bar{P}_{ESS,n}^{\text{ch}}} + \frac{p_{ESS,n,t}^{\text{dc}}}{\bar{P}_{ESS,n}^{\text{dc}}} \leq 1, \forall t, \quad (\lambda_{ESS,n,t}^{\text{relax}}) \quad (16)$$

where $\lambda_{ESS,n,t}^{\text{relax}}$ is the dual multiplier.

Generally, the complementarity constraint of an ESS is introduced to avoid SCD, as illustrated by Fig. 2 [9]:

$$p_{ESS,n,t}^{\text{dc}} \cdot p_{ESS,n,t}^{\text{ch}} = 0 \quad (17)$$

However, complementary constraints make the optimization problem non-convex and challenging to solve. To cope with this issue, binary variables are usually employed to transform the complementarity constraints into mixed-integer programming [10].

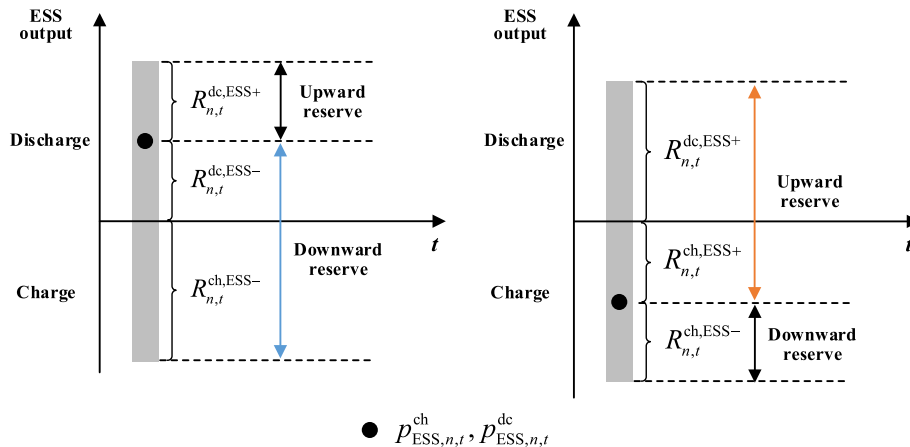


Fig. 1. Illustration of the downward/upward reserves when ESSs are in the charging/discharging status.

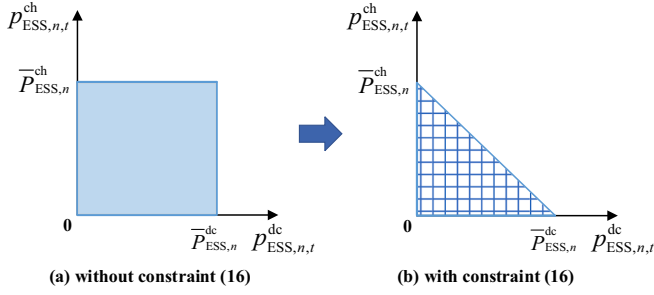


Fig. 2. Illustration of the feasible region of the charging and discharging active power of the n^{th} ESS at time t with or without constraint (16).

$$\begin{aligned}
 0 &\leq p_{\text{ESS},n,t}^{\text{ch}} \leq z_{\text{ESS},n,t}^{\text{ch}} \cdot \bar{P}_{\text{ESS},n}^{\text{ch}} \\
 0 &\leq p_{\text{ESS},n,t}^{\text{dc}} \leq z_{\text{ESS},n,t}^{\text{dc}} \cdot \bar{P}_{\text{ESS},n}^{\text{dc}} \\
 z_{\text{ESS},n,t}^{\text{ch}} + z_{\text{ESS},n,t}^{\text{dc}} &\leq 1 \\
 z_{\text{ESS},n,t}^{\text{ch}}, z_{\text{ESS},n,t}^{\text{dc}} &\in \{0, 1\}
 \end{aligned} \quad (18)$$

Nevertheless, the MIP remains computationally complex as the number of ESSs or the time periods increases in multi-period optimization problems [8]. Subsequently, we will develop an exact generalized relaxation condition that provably ensure no SCD in the “**relaxed model**”, that is, the relaxed optimization problems excluding the complementarity constraints (17).

Remark 1. It is noteworthy that the proposed relaxation method is applicable to all types of energy storage that require the avoidance of simultaneous charging and discharging, including but not limited to electrochemical energy storage, pumped hydro storage and compressed air energy storage.

3. Exact relaxation condition for complementarity constraints of storages

In this section, we first present the proposed GRC for exactly relaxing ESSs' complementarity constraints. Then, discussions regarding the inclusion relationships between the proposed GRC and the other existing ones are given, accompanied by theoretical proofs.

3.1. The proposed exact relaxation condition for relaxing ESSs' complementarity constraints

The proposed relaxation condition to avoid SCD in the relaxed model is given as follows:

Condition C1: we say that Condition C1 holds when the following inequality holds:

$$\lambda_{j,t}^{\text{p}} > \lambda_{j,t}^{\text{p,C1}} \quad (19)$$

where $\lambda_{j,t}^{\text{p}}$ is the multiplier of the active power balance constraint at bus j and time t , that is, the LMP. The expression of $\lambda_{j,t}^{\text{p,C1}}$ is given as follows:

$$\lambda_{j,t}^{\text{p,C1}} = \left[\frac{\partial \text{obj}}{\partial (p_{\text{ESS},n,t}^{\text{ch}})} \eta_{\text{ESS},n}^{\text{dc}} + \frac{\partial \text{obj}}{\partial (p_{\text{ESS},n,t}^{\text{dc}})} \eta_{\text{ESS},n}^{\text{ch}} \right] / \left(\eta_{\text{ESS},n}^{\text{ch}} - \frac{1}{\eta_{\text{ESS},n}^{\text{dc}}} \right) \quad (20)$$

Theorem 1. as long as the Condition C1 is satisfied, the relaxation is guaranteed to be exact, that is, SCD automatically holds.

Proof: The detailed proof is given in Appendix A. ■

Remark 2. Actually, the proposed relaxation condition does not depend on the convexity of the power grid optimization problems. The following outlines the detailed analytical process:

- i) Based on the proof process detailed in Appendix A, it becomes apparent that $p_{\text{ESS},n,t}^{\text{ch}} \cdot p_{\text{ESS},n,t}^{\text{dc}} = 0$ can be derived combining the proposed relaxation condition with the KKT conditions. In other words, based on the proposed relaxation condition, the (local) optimal solution enforces the complementary constraint, even though it is relaxed, as illustrated by the light coral box in Fig. 3.
- ii) Subsequently, combining *Assumption i)*, we can conclude that the (local) optimal solution for problem (P1) satisfies the KKT conditions. *Assumption ii)* serves to guarantee the validity of *Assumption i)*, as shown by the light purple box in Fig. 3. In other words, the KKT conditions are necessary conditions for the (local) optimal solution of optimization problem (P1) only if the constraint qualifications are satisfied.
- iii) Based on the above two points of analysis, we can conclude that for power grid optimization problems satisfying *Assumptions i)* and *ii)*, it can be established that at their (local) optimal solution, the proposed relaxation condition ensures the exactness of the relaxation, as demonstrated by the light green box in Fig. 3.

Remark 3. The proposed relaxation condition is independent from the forms of objective functions and operational constraints, as indicated by the derivation process. Thus, this condition is applicable for many applications, such as storage- participating ED, joint energy-reserve optimization and Volt/VAR optimization.

Fig. 4 presents the visualization of **Condition C1**. We denote the range characterized by (19) as the **exact relaxation region** of $\lambda_{j,t}^{\text{p}}$. Besides, $\lambda_{j,t}^{\text{p,C1}}$ given in (20) is referred to as the **lower boundary on LMP** of the proposed relaxation condition. As long as the LMP is higher than the lower boundary, that is, the **Condition C1** holds, the relaxation is guaranteed to be exact.

3.2. The inclusion relationships between the proposed relaxation condition and the other existing ones

In this subsection, we discuss the inclusion relationships between the proposed relaxation condition and the other existing ones in [20–24]. Although the conditions given in [20–22] are based upon the ED model, similar relaxation conditions can also be derived for the established optimization model, which are elaborated in Appendix B.

Lemma 1. The exact relaxation region about LMPs formulated by the proposed relaxation condition (**Condition C1**) contains those of the following relaxation conditions:

- i) the Relaxation conditions in [20] (i.e., the Relaxation conditions of group A in [21]);
- ii) the Relaxation conditions of group B in [21];
- iii) the Relaxation conditions of group C in [21];

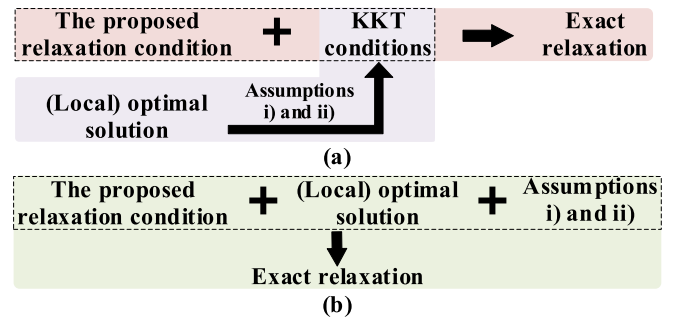


Fig. 3. Illustration of the relationship between the proposed relaxation condition and the assumptions of the optimization model.

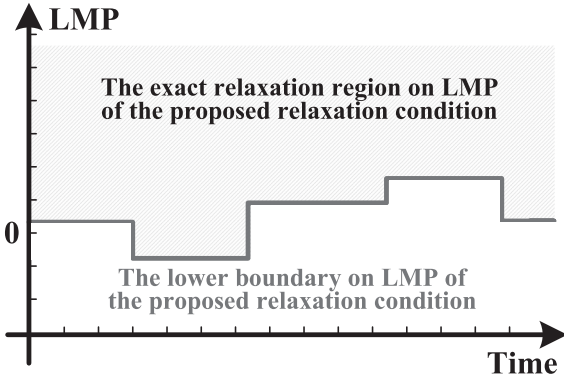


Fig. 4. Illustration of the exact relaxation region and the lower boundary on LMP of the proposed relaxation condition.

- iv) the Relaxation conditions in [22];
- v) the Relaxation conditions in [23];
- vi) the Relaxation conditions in [24].

Proof: The detailed proofs are given in Appendix C. We also give elaborate descriptions of these relaxation conditions in Appendixes B and C to facilitate the proofs. ■

Remark 4. Lemma 1 demonstrates that the proposed relaxation condition can achieve a broader exact relaxation region than existing relaxation conditions in [20–24]. Correspondingly, Fig. 6 illustrates the inclusion relationships between different relaxation conditions given in Lemma 1. Due to space limitations, refer to Appendix F in the supplementary file [31] for detailed demonstrations. Moreover, according to the proofs in Appendix C, we can also conclude that the presented Lemma 1 does not rely on the specific objective forms and operational constraints, and thus it is generally valid in various scenarios, such as storage- participating ED, joint energy-reserve optimization and Volt/VAR optimization. As far as we know, it is the first time that the inclusion relationships among multiple energy storage relaxation conditions are established.

Remark 5. The inclusion relationship mentioned here actually refers to those of the exact LMP relaxation regions corresponding to different relaxation conditions. Fig. 5 gives an illustration, wherein the gray corresponds to the proposed relaxation condition, and the aqua is related to the relaxation condition in [20]. As has been demonstrated in Lemma 1, the exact LMP relaxation region corresponding to the proposed relaxation condition includes that in [20]. In addition, a visual illustration of the exact relaxation range under different parameters is

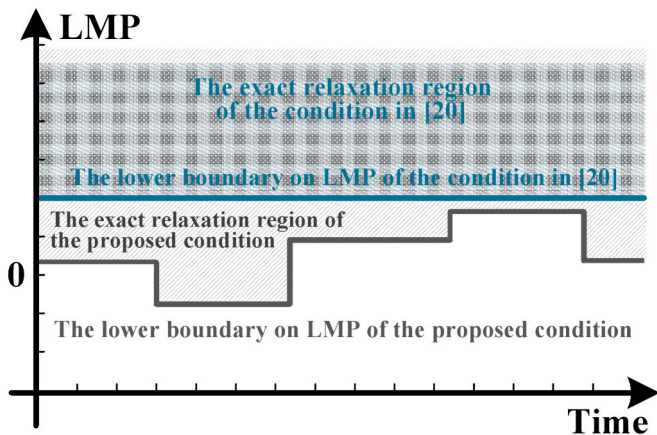


Fig. 5. Illustration of the inclusion relationships between the proposed relaxation condition and that in [20].

provided in Subsection E of Section V.

Remark 6. To elaborate on the applicability of the proposed relaxation condition across various scenarios, we also analyze the proposed condition for ESSs under multi-period ED scenario. The details are given in Appendix E.

4. Practical relaxation method for complementarity constraints of storages

The proposed GRC can generally be met in most electricity price scenarios, including positive prices and moderate negative prices. While notice that scenarios with extreme negative prices may still occur occasionally. To guarantee the exactness of relaxation in all electricity price scenarios, we further introduce penalties for charging and discharging losses of storage into the objective function, and give details on their benefits to the storage relaxation. Besides, we establish a criterion for appropriately determining the penalty parameter that ensures the satisfaction of the GRC. Afterward, a practical relaxation method is presented based on the proposed GRC, which does not heavily rely on the prediction accuracy of LMP.

Prior to presenting the practical relaxation method, we first explicitly express the terms regarding the charging and discharging active power of ESSs in the objective function of problem (P1). Specifically, these terms include: i) the penalty for the charging and discharging losses, i.e., formula (24); ii) the energy storage arbitrage term (25). Then, we can obtain the following problem (P2):

$$\begin{aligned} \min_{\mathbf{x}} \text{obj} = & C(\mathbf{x}) + \underbrace{C_{\text{ESS}}^{\text{Pen}}}_{\text{penalty for the charging and discharge losses}} \\ & + \underbrace{\sum_{t=1}^T \sum_{n \in \Phi_{\text{ESS}}} [f_{n,t}(p_{\text{ESS},n,t}^{\text{ch}}) + g_{n,t}(p_{\text{ESS},n,t}^{\text{dc}})]}_{\text{energy storage arbitrage}} \end{aligned} \quad (21)$$

$$s.t. G(\mathbf{x}) \leq 0 \quad (22)$$

$$H(\mathbf{x}) = 0 \quad (23)$$

where the aforementioned inequality constraints (6)–(14), and (16) are embodied in (22). Similarly, formulas (4)–(5), and (15) are included in (23).

The specific expressions of $C_{\text{ESS}}^{\text{Pen}}, f_{n,t}, g_{n,t}$ are given as follows:

$$C_{\text{ESS}}^{\text{Pen}} = \sigma_{\text{ESS}} \sum_{t=1}^T \sum_{n \in \Phi_{\text{ESS}}} \left\{ p_{\text{ESS},n,t}^{\text{dc}} \left(\frac{1}{\eta_{\text{ESS},n}^{\text{dc}}} - 1 \right) + p_{\text{ESS},n,t}^{\text{ch}} \left(1 - \eta_{\text{ESS},n}^{\text{ch}} \right) \right\} \quad (24)$$

$$\begin{aligned} f_{n,t}(p_{\text{ESS},n,t}^{\text{ch}}) &= a_t^{\text{ch}} \cdot p_{\text{ESS},n,t}^{\text{ch}} \\ g_{n,t}(p_{\text{ESS},n,t}^{\text{dc}}) &= a_t^{\text{dc}} \cdot p_{\text{ESS},n,t}^{\text{dc}} \end{aligned} \quad (25)$$

where $C_{\text{ESS}}^{\text{Pen}}$ is the penalty term for the charging and discharging losses of storages [19]. $f_{n,t}$ and $g_{n,t}$ are the charging and discharging fees of the n^{th} ESS at time t . σ_{ESS} is the penalty parameter. T is the whole optimization period. Φ_{ESS} is the index set of ESSs. a_t^{ch} and a_t^{dc} are the charging and discharging prices at time t .

Remark 7. Given that $\eta_{\text{ESS},n}^{\text{dc}}$ and $\eta_{\text{ESS},n}^{\text{ch}}$ are both positive numbers less than one, a penalty is triggered as long as either $p_{\text{ESS},n,t}^{\text{dc}}$ or $p_{\text{ESS},n,t}^{\text{ch}}$ is non-zero.

Remark 8. Note that there have been many studies characterize the energy storage arbitrage as a linear formulation, such as those objective functions in [16,20,21,32,33]. Therefore, we also adopt the same formulation (25).

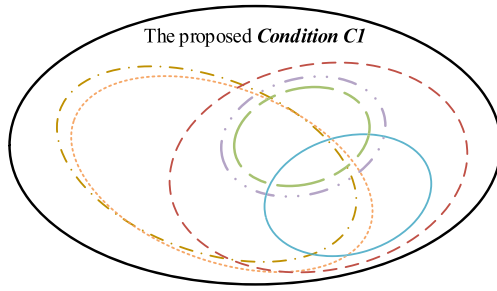


Fig. 6. The inclusion relationships of the LMP relaxation regions corresponding to different relaxation conditions.

4.1. The criterion for determining the penalty parameter σ_{ESS} to ensure the satisfaction of the proposed GRC

In this subsection, we reformulate the proposed GRC for problem (P2). Considering the objective function (21), the **Condition C1** can be rearranged as:

$$\begin{aligned} \lambda_{j,t}^{\text{p,C1}} &= \left[\frac{\partial \text{obj}}{\partial (p_{\text{ESS},n,t}^{\text{ch}})} \Big/ \eta_{\text{ESS},n}^{\text{dc}} + \frac{\partial \text{obj}}{\partial (p_{\text{ESS},n,t}^{\text{dc}})} \eta_{\text{ESS},n}^{\text{ch}} \right] \Big/ \left(\eta_{\text{ESS},n}^{\text{ch}} - \frac{1}{\eta_{\text{ESS},n}^{\text{dc}}} \right) \\ &= \left[\frac{\partial (f_{n,t})}{\partial (p_{\text{ESS},n,t}^{\text{ch}})} \Big/ \eta_{\text{ESS},n}^{\text{dc}} + \frac{\partial (g_{n,t})}{\partial (p_{\text{ESS},n,t}^{\text{dc}})} \eta_{\text{ESS},n}^{\text{ch}} \right] \Big/ \left(\eta_{\text{ESS},n}^{\text{ch}} - \frac{1}{\eta_{\text{ESS},n}^{\text{dc}}} \right) \\ &+ \left[\frac{\partial (C_{\text{ESS}}^{\text{pen}})}{\partial (p_{\text{ESS},n,t}^{\text{ch}})} \Big/ \eta_{\text{ESS},n}^{\text{dc}} + \frac{\partial (C_{\text{ESS}}^{\text{pen}})}{\partial (p_{\text{ESS},n,t}^{\text{dc}})} \eta_{\text{ESS},n}^{\text{ch}} \right] \Big/ \left(\eta_{\text{ESS},n}^{\text{ch}} - \frac{1}{\eta_{\text{ESS},n}^{\text{dc}}} \right) \end{aligned} \quad (26)$$

Substitute (25) into above formula, we have

$$\begin{aligned} &\left[\frac{\partial (f_{n,t})}{\partial (p_{\text{ESS},n,t}^{\text{ch}})} \Big/ \eta_{\text{ESS},n}^{\text{dc}} + \frac{\partial (g_{n,t})}{\partial (p_{\text{ESS},n,t}^{\text{dc}})} \eta_{\text{ESS},n}^{\text{ch}} \right] \Big/ \left(\eta_{\text{ESS},n}^{\text{ch}} - \frac{1}{\eta_{\text{ESS},n}^{\text{dc}}} \right) \\ &= \left[a_t^{\text{ch}} / \eta_{\text{ESS},n}^{\text{dc}} + a_t^{\text{dc}} \cdot \eta_{\text{ESS},n}^{\text{ch}} \right] \Big/ \left(\eta_{\text{ESS},n}^{\text{ch}} - 1 / \eta_{\text{ESS},n}^{\text{dc}} \right) \end{aligned} \quad (27)$$

Combining eq. (24), we can obtain

$$\begin{aligned} &\left[\frac{\partial (C_{\text{ESS}}^{\text{pen}})}{\partial (p_{\text{ESS},n,t}^{\text{ch}})} \Big/ \eta_{\text{ESS},n}^{\text{dc}} + \frac{\partial (C_{\text{ESS}}^{\text{pen}})}{\partial (p_{\text{ESS},n,t}^{\text{dc}})} \eta_{\text{ESS},n}^{\text{ch}} \right] \Big/ \left(\eta_{\text{ESS},n}^{\text{ch}} - \frac{1}{\eta_{\text{ESS},n}^{\text{dc}}} \right) \\ &= \sigma_{\text{ESS}} \left[\left(1 - \eta_{\text{ESS},n}^{\text{ch}} \right) \Big/ \eta_{\text{ESS},n}^{\text{dc}} + \left(\frac{1}{\eta_{\text{ESS},n}^{\text{dc}}} - 1 \right) \cdot \eta_{\text{ESS},n}^{\text{ch}} \right] \Big/ \left(\eta_{\text{ESS},n}^{\text{ch}} - \frac{1}{\eta_{\text{ESS},n}^{\text{dc}}} \right) \\ &= \sigma_{\text{ESS}} \left[\frac{1 - \eta_{\text{ESS},n}^{\text{ch}} + \eta_{\text{ESS},n}^{\text{ch}} \eta_{\text{ESS},n}^{\text{dc}} - \eta_{\text{ESS},n}^{\text{dc}} \eta_{\text{ESS},n}^{\text{ch}}}{\eta_{\text{ESS},n}^{\text{dc}}} \right] \Big/ \left(\eta_{\text{ESS},n}^{\text{ch}} - \frac{1}{\eta_{\text{ESS},n}^{\text{dc}}} \right) \\ &= \sigma_{\text{ESS}} \left[\frac{1 - \eta_{\text{ESS},n}^{\text{dc}} \eta_{\text{ESS},n}^{\text{ch}}}{\eta_{\text{ESS},n}^{\text{dc}}} \right] \Big/ \left(\eta_{\text{ESS},n}^{\text{ch}} - \frac{1}{\eta_{\text{ESS},n}^{\text{dc}}} \right) \\ &= -\sigma_{\text{ESS}} \end{aligned} \quad (28)$$

Then, the **Condition C1** can be reformulated as follows by combining (19) and (26)–(28).

$$\begin{aligned} \lambda_{j,t}^{\text{p}} &> \lambda_{j,t}^{\text{p,C1}} \\ &= -\sigma_{\text{ESS}} + \left[a_t^{\text{ch}} / \eta_{\text{ESS},n}^{\text{dc}} + a_t^{\text{dc}} \cdot \eta_{\text{ESS},n}^{\text{ch}} \right] \Big/ \left(\eta_{\text{ESS},n}^{\text{ch}} - 1 / \eta_{\text{ESS},n}^{\text{dc}} \right) \end{aligned} \quad (29)$$

Therefore, we have following criterion to set the penalty parameter value:

$$\sigma_{\text{ESS}} > -\lambda_{j,t}^{\text{p}} + \left[a_t^{\text{ch}} / \eta_{\text{ESS},n}^{\text{dc}} + a_t^{\text{dc}} \cdot \eta_{\text{ESS},n}^{\text{ch}} \right] \Big/ \left(\eta_{\text{ESS},n}^{\text{ch}} - 1 / \eta_{\text{ESS},n}^{\text{dc}} \right) \quad (30)$$

Obviously, as long as σ_{ESS} is large enough to make (30) holds, the

- i) the Relaxation conditions in [20] (i.e. the Relaxation conditions of group A in [21])
- ii) the Relaxation conditions of group B in [21]
- iii) the Relaxation conditions of group C in [21]
- · - iv) the Relaxation conditions in [22]
- v) the Relaxation conditions in [23]
- vi) the Relaxation conditions in [24]

exactness of the energy storage relaxation can be ensured in all scenarios even for those with extreme negative prices. This characteristic highlights the benefit of penalizing storage charging and discharging losses for the exact relaxation of energy storages.

Discussion 1: Some discussions about (29) are given as follows:

1) The parameters σ_{ESS} , a_t^{ch} , a_t^{dc} , $\eta_{\text{ESS},n}^{\text{ch}}$, $\eta_{\text{ESS},n}^{\text{dc}}$ are all known constants, thus the right-hand term of formula (29) is a constant. This makes the proposed GRC exhibits a priori property, which facilitates practical applications.

2) Parameters $\eta_{\text{ESS},n}^{\text{ch}}$ and $\eta_{\text{ESS},n}^{\text{dc}}$ are both slightly less than 1, indicating that $(\eta_{\text{ESS},n}^{\text{ch}} - 1 / \eta_{\text{ESS},n}^{\text{dc}})$ is negative and slightly smaller than zero. If $[a_t^{\text{ch}} / \eta_{\text{ESS},n}^{\text{dc}} + a_t^{\text{dc}} \cdot \eta_{\text{ESS},n}^{\text{ch}}]$ is positive, then $[a_t^{\text{ch}} / \eta_{\text{ESS},n}^{\text{dc}} + a_t^{\text{dc}} \cdot \eta_{\text{ESS},n}^{\text{ch}}] / (\eta_{\text{ESS},n}^{\text{ch}} - 1 / \eta_{\text{ESS},n}^{\text{dc}})$ is a small enough negative value. This situation is usually satisfied in real scenarios, which partly explains why the proposed relaxation condition can ensure exactness even under negative prices.

4.2. The practical relaxation method

Given a priori characteristic of the proposed relaxation condition, a practical relaxation method is presented based on the proposed GRC:

S1) Forecast LMPs based on the historical data of LMPs, load forecast, renewable forecast and ancillary service requirements, etc. [34]. Combining the degree of accuracy for historical forecasts, set appropriate mean absolute percentage error (MAPE) as an approximate estimation of the standard deviation in LMP forecasting. Then, in order to minimize the impact of prediction errors on the proposed GRC, we utilize the lower bound of the interval into which the actual LMP may fall, denoted as $\underline{\text{LMP}}_{j,t}$, rather than $\text{LMP}_{j,t}$, for subsequent calculation.

$$\underline{\text{LMP}}_{j,t} = \begin{cases} (1 - 5 \times \text{MAPE}) \times \text{LMP}_{j,t}^{\text{pred}}, & \text{if } \text{LMP}_{j,t}^{\text{pred}} > 0 \\ (1 + 5 \times \text{MAPE}) \times \text{LMP}_{j,t}^{\text{pred}}, & \text{otherwise} \end{cases} \quad (31)$$

where $\text{LMP}_{j,t}^{\text{pred}}$ is the predicted LMP.

S2) Input charging/discharging prices a_t^{ch} , a_t^{dc} and charging/discharging efficiencies $\eta_{\text{ESS},n}^{\text{ch}}$, $\eta_{\text{ESS},n}^{\text{dc}}$.

S3) Set the value of penalty parameter σ_{ESS} to make (32) holds.

$$\sigma_{\text{ESS}} > -\underline{\text{LMP}}_{j,t} + \left[a_t^{\text{ch}} / \eta_{\text{ESS},n}^{\text{dc}} + a_t^{\text{dc}} \cdot \eta_{\text{ESS},n}^{\text{ch}} \right] \Big/ \left(\eta_{\text{ESS},n}^{\text{ch}} - 1 / \eta_{\text{ESS},n}^{\text{dc}} \right) \quad (32)$$

Remark 9. As a matter of fact, using $\underline{\text{LMP}}_{j,t}$ instead of $\text{LMP}_{j,t}$ facilitates the applicability of the proposed relaxation method. This is because predicting the range within which the LMPs are likely to lie, is easier to implement than accurately determining the actual LMP values in practical scenarios.

Remark 10. It is essential to emphasize that, despite we utilize the LMP predictions, the **practical relaxation method does not heavily rely on the accuracy of LMP predictions**. The reasons are twofold: on

the one hand, we employ $\underline{\text{LMP}}_{j,t}$, rather than $\text{LMP}_{j,t}$, shown in (31), indicating that there is no need to precisely determine the actual LMP values; on the other hand, the penalty parameter σ_{ESS} in (24) can be set appropriately larger. Both of these measures can provide some reserved margins to ensure the exactness of the relaxation, as illustrated in Fig. 7.

For ease of understanding, we provide a small example where $a_t^{\text{ch}} = -7$ \$/MWh, $a_t^{\text{dc}} = 15$ \$/MWh, $\eta_{\text{ESS},n}^{\text{ch}} = \eta_{\text{ESS},n}^{\text{dc}} = 0.93$, $\text{LMP}_{j,t}^{\text{pred}} = -30$ \$/MWh. To highlight that the practical relaxation method indeed does not heavily rely on the accuracy of LMP predictions, here we set a relatively large standard deviation in LMP forecasting. Specifically, MAPE is set as 8% in (31). Then, combining (31), we can obtain $\underline{\text{LMP}}_{j,t} = -42$ \$/MWh. Furthermore, we have $[a_t^{\text{ch}}/\eta_{\text{ESS},n}^{\text{dc}} + a_t^{\text{dc}} \cdot \eta_{\text{ESS},n}^{\text{ch}}] / (\eta_{\text{ESS},n}^{\text{ch}} - 1/\eta_{\text{ESS},n}^{\text{dc}}) = -44.22$ \$/MWh. Moreover, we set $\sigma_{\text{ESS}} = 3$. In this case, $\lambda_{j,t}^{\text{p,C1}} = -47.22$ \$/MWh. Then, it is easy to obtain $\lambda_{j,t}^{\text{p}} \geq -42$ \$/MWh $>$ $\lambda_{j,t}^{\text{p,C1}} = -47.22$ \$/MWh, ensuring the exactness of the relaxation. Wherein the reserved margin is $-30 - (-47.22) = 17.22$ \$/MWh.

For occasional scenarios where there are significant deviations between the predicted LMPs and their real values, we further propose a practical two-stage penalty parameter adjustment scheme. By increasing the penalty parameter σ_{ESS} to satisfy formula (30), we can ensure the relaxation exactness. The specific implementation procedure is presented in Fig. 8:

5. Numerical Tests

As one of the typical application scenarios, we consider the multi-period active and reactive power joint coordination optimization problem here, which considers energy and reserve services simultaneously. The specific form of the 24-h optimization problem is detailed in Appendix D. We then conducted case studies in multiple power systems of different scales to validate the advantages and effectiveness of the proposed exact relaxation condition.

The charging/discharging efficiencies of storages are randomly generated within the range of [0.9, 0.93]. The detailed parameters about loads, ESSs, RGs, generators are given in [35]. Moreover, we set self-discharge rate as 0.002, $\sigma_{\text{ESS}} = 3$, $\Delta t = 1$ h. The upward/downward reserve prices of generators are set as $c_{g,t}^{G+} = c_{g,t}^{G-} = 10$ \$/MWh, and the upward/downward reserve prices of storages are set as $c_{n,t}^{\text{ESS}+} = c_{n,t}^{\text{ESS}-} = 5$ \$/MWh. The solver for continuous optimization problems is IPOPT [36], and BONMIN [37] for MIP. Case studies are conducted in Matlab 2019b on a laptop with an Intel i7-10875H CPU and 24GB RAM.

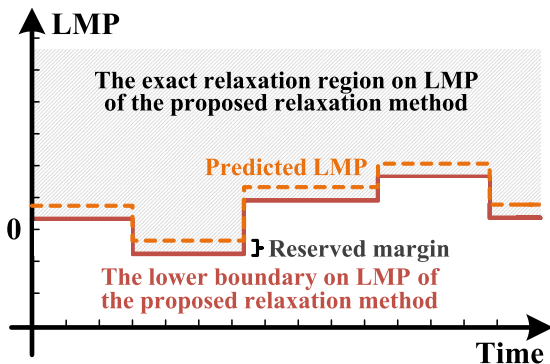


Fig. 7. Illustration of the predicted LMP, the lower boundary of LMP and the reserved margin.

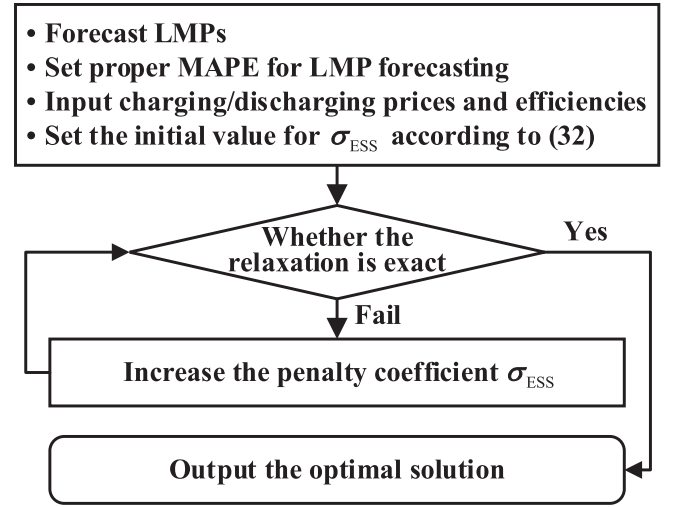


Fig. 8. Flowchart of the practical two-stage penalty parameter adjustment scheme.

5.1. Comparing the computational efficiency of the relaxed model and the MIP model across multiple cases

Table 3 presents the computation time comparison between the relaxed model and the MIP model within several power systems of different scales. It can be observed that:

- The optimal objective function values of the relaxed model and the MIP model are identical across several system scales.
- As the system scale and the number of storage devices increase, the computation time of the MIP model increases rapidly, while the solution time of the relaxed model optimization problem grows relatively slowly. The advantage becomes more significant with the enlargement of the system scale and the rise in the number of storage devices, highlighting the necessity of relaxing complementarity constraints for storages.

5.2. The day-ahead case with negative LMP

Subsequently, taking the IEEE 69-bus system as an example, some numerical tests are given under day-ahead negative price scenarios. Fig. 9 illustrates the total loads and available renewable energy. It can be observed that the net load values corresponding to the gray areas are negative.

Given that $a_t^{\text{dc}} = a_t^{\text{ch}} = 15$ \$/MWh and the renewable generation cost

Table 3

Comparing the Computational Efficiency of the Relaxed Model and the MIP Model Across Multiple Cases.

Case	IEEE 69	IEEE 118	IEEE 300
Charging price (\$/MWh)	-10	-10	-10
Discharging price (\$/MWh)	15	15	15
Whether the Condition C1 is met	yes	yes	yes
Whether the relaxed model is accurate	yes	yes	yes
Optimal value of the relaxed model (\$)	3807.4	2.839×10^6	3.641×10^7
Optimal value of the MIP model (\$)	3807.4	2.839×10^6	3.641×10^7
Solving time of the relaxed model (s)	12.35	68.85	85.62
Computation time of the MIP model (s)	64.93	509.48	916.13
Number of binary variables in the MIP model	1680	2592	6816

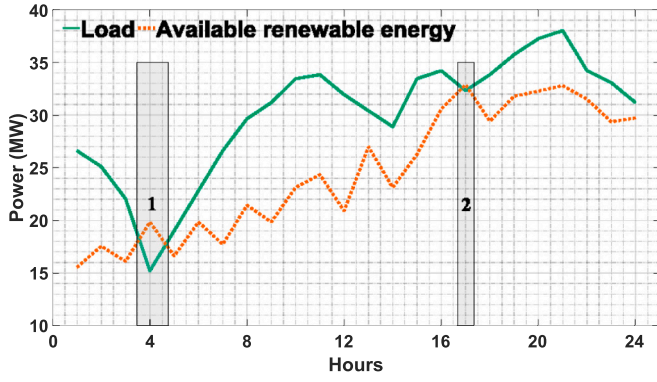
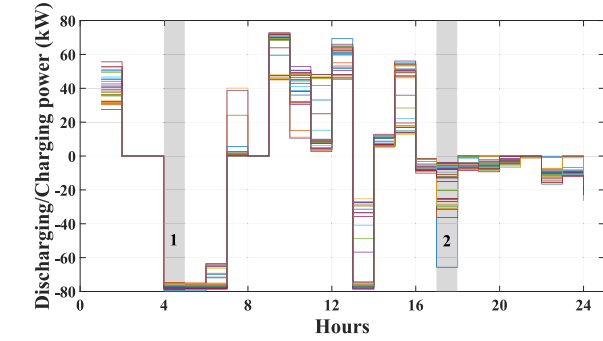
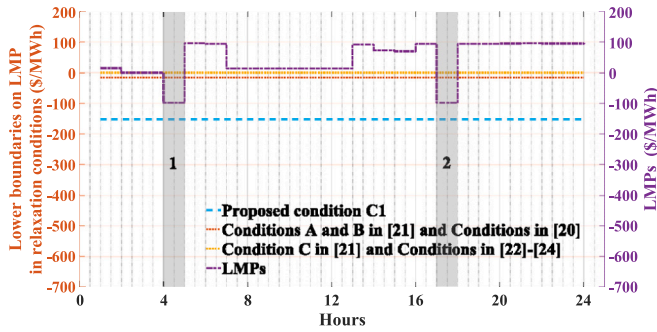


Fig. 9. Illustration of the total loads and available renewable energy in IEEE 69-bus system.

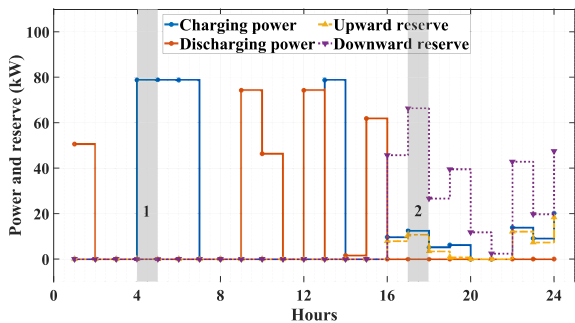
is -100 \$/MWh, representing scenarios with notably high penetrations of renewable energy or transmission congestion, the optimization results are depicted in Fig. 10. In subfigure (a), the gray areas indicate that



(a) Charging/discharging power of storages
(Discharging corresponds to positive values, while negative values for charging)



(b) Verifying the validity of different relaxation conditions



(c) Upward/downward reserves supplied by the storage at bus 10

Fig. 10. The results employing the relaxed model under negative price scenarios.

storage charging takes place during periods when the available renewable energy exceeds the load demand, as illustrated in Fig. 9. This is because storages absorbing surplus power helps minimize the renewable energy curtailment.

As shown in subfigure (b), the results not only demonstrate that *Condition C1* holds in this scenario, but also verify that *Lemma 1* is satisfied. Furthermore, the negative LMPs appear exactly during the periods when the available renewable energy exceeds the load demand, that is, the gray areas. Correspondingly, it can be observed that there is no SCD in the optimization results of subfigure (a), verifying the exactness of the proposed condition. In contrast, the relaxation conditions of Group C in [21] and conditions in [22–24] do not hold due to the existence of negative LMPs. As for the relaxation conditions of groups A and B in [21] and conditions in [20], none of them hold since the negative price values are quite small, illustrating the advantages of the proposed condition under negative prices.

In addition, subfigure (c) presents the upward/downward reserves supplied by the storage at bus 10. Take the 2nd gray area as an example, it's worth noting that ESSs can provide upward reserves by reducing the charging power. Similarly, downward reserve capacity can be offered by charging more from their predetermined power schedules.

5.3. Verifying the exactness of the proposed relaxation condition under multiple scenarios

Hereafter, if not specifically stated, all tests are carried out in the IEEE 69-bus system. Moreover, in the following scenarios, we assume that the renewable generation cost is -5 \$/MWh. Then, we compare the results obtained by the relaxed model with those of the MIP model. As presented in Table 4, the relaxed model can obtain consistent energy storage charging/discharging power and objective values as those of the MIP model in all six scenarios, which justifies the validity of the proposed relaxation condition. The results also show that the relaxed model can be calculated much more efficiently than the original MIP model for various scenarios.

Moreover, extensive numerical tests have been conducted across multiple application scenarios, system scales, and charging/discharging prices to verify the effectiveness and robustness of the proposed relaxation condition. Due to space limitations, refer to Appendix G in the supplementary file [31] for details.

Table 4

Verifying the Exactness and Efficiency of the Proposed Relaxation Condition Under Different Scenarios.

Scenario	S1	S2	S3	S4	S5	S6
Charging price (\$/MWh)	0	-15	-10	0	5	10
Discharging price (\$/MWh)	0	20	15	15	15	15
Whether the Condition C1 is met	yes	yes	yes	yes	yes	yes
Whether the relaxed model is accurate	yes	yes	yes	yes	yes	yes
Optimal value of the relaxed model (\$)	3776.8	3790.4	3807.4	3941.6	3991.3	4034.6
Optimal value of the MIP model (\$)	3776.8	3790.4	3807.4	3941.6	3991.3	4034.6
Solving time of the relaxed model (s)	14.83	12.75	12.35	14.61	11.80	11.82
Calculation time of the MIP model (s)	75.48	68.74	64.93	218.82	164.57	63.21

Remark 11. According to the combinations of signs (positive or negative) for both discharging price a_t^{dc} and charging price a_t^{ch} , these scenarios presented in Table 5 can be further divided into the following three categories:

- 1) In category 1, the costs related to charging and discharging are omitted, as discussed in [8,19,38,39]. This situation may happen when the storage device is owned by the power grid company itself.
- 2) Category 2, which has been extensively considered in the literature [20–24], involves the storage owner paying the grid for charging energy while receiving compensation from the grid for discharging. To incentivize owners to discharge stored energy back to the grid, the marginal compensation offered for discharging one unit of energy should cover the owner's marginal charging cost for the equivalent amount of energy, i.e. $a_t^{ch} + a_t^{dc} > 0$.
- 3) Category 3 is likely to occur when storages facilitate the absorption of excess renewable energy and help to maintain the grid's energy balance. In this category, the grid incentivizes storage charging. Note that such a situation may occur in practice but not necessarily in every period.

5.4. Verifying the validity of different relaxation conditions under negative LMP scenarios

Under different charging/discharging prices and renewable generation costs, we verify whether the proposed relaxation condition and other conditions in [8,20–24] are valid. The results are presented in Table 6. The results verify that the proposed relaxation condition still holds in scenarios even with a very negative LMP, as shown in scenarios S7–S11. Therefore, we can draw the conclusion that as long as the proposed relaxation condition hold, it can be guaranteed that there is no SCD. In contrast, those relaxation conditions given in [8,20–24] do not hold in these scenarios, which highlights the advantages of the proposed condition. Moreover, the results also demonstrate that the proposed relaxation condition has a broader application range compared with the other existing conditions.

Additionally, comprehensive numerical tests have been carried out across multiple application scenarios, system scales, charging/discharging prices and renewable generation costs to validate the superiority of the proposed relaxation condition. Due to space limitations, refer to Appendix I in the supplementary file [31] for details.

5.5. Illustrating the exact relaxation region and verifying the effectiveness of the practical relaxation method

During periods when the available renewable energy exceeds the load demand, renewable energy producers tend to offer electricity at negative prices to promote the integration of renewable energy. Nevertheless, in these scenarios, $LMP_{j,t}$ will be significantly influenced

Table 5

Categorizing Application Scenarios Where the Proposed Relaxation Condition can be Applied.

Category	Description	Signs of (a_t^{dc} , a_t^{ch})	Covered scenarios
Category 1	The operational costs of ESSs are disregarded in the grid's dispatch at any time t	$a_t^{dc} = 0$ $a_t^{ch} = 0$	S1
Category 2	The storages' owner pays the grid for the charging energy, and in turn, the grid compensates the ESS owner for the discharging energy at time t	$a_t^{dc} > 0$ $a_t^{ch} < 0$	S2, S3
Category 3	The grid pays the ESS owner for both charging and discharging energy at time t	$a_t^{dc} > 0$ $a_t^{ch} > 0$	S5, S6

Table 6

Verifying the Validity of Different Relaxation Conditions Under Negative LMP Scenarios.

Scenario	S7	S8	S9	S10	S11
Charging price (\$/MWh)	−8	−5	−3	15	20
Discharging price (\$/MWh)	15	15	15	15	15
Renewable generation cost (\$/MWh)	−10	−20	−30	−100	−120
Minimal LMP (\$/MWh)	−10.67	−20.53	−30.28	−99.1	−118.8
Whether the conditions in [8] are met	no	no	no	no	no
Whether the conditions in [20] are met	no	no	no	no	no
Whether the conditions of group B in [21] are met	no	no	no	no	no
Whether the conditions of group C in [21] are met	no	no	no	no	no
Whether the conditions in [22] are met	no	no	no	no	no
Whether the conditions in [23] are met	no	no	no	no	no
Whether the conditions in [24] are met	no	no	no	no	no
Whether Condition C1 is met	yes	yes	yes	yes	yes
Whether the relaxed model is accurate	yes	yes	yes	yes	yes

by the cost coefficient of renewable energy. Moreover, considering that the derived practical relaxation method is related to the charging and discharging prices, thus the exact relaxation range for storages under varying charging price and renewable energy cost coefficient is presented below, as illustrated in Fig. 11. For the sake of illustration, we unify the charging price and the cost coefficient for renewable energy. As for the discharging price, it is set as 15 \$/MWh. The results demonstrate that:

- 1) The exact relaxation region obtained based on the proposed Condition C1 is much larger than the others in [20,21]. As for those conditions in [22–24], they are all included in the relaxation conditions of group C in [21], as demonstrated by the proofs in Appendix C. Therefore, we can conclude that the exact relaxation region about LMPs outlined by the proposed Condition C1 includes those described in [20–24].
- 2) In this figure, the solid red line and brown line are the lower boundaries of the exact relaxation regions correspond to the cases of $\sigma_{ESS} = 3$ and $\sigma_{ESS} = 30$, respectively. It can be found that the lower boundary of the exact relaxation region gradually shifts downward as σ_{ESS} increases, that is, an increase in the penalty parameter σ_{ESS} leads to a broader relaxation range.

Take the purple upper triangle as an example, it is outside the exact relaxation region corresponding to $\sigma_{ESS} = 3$, but inside the exact relaxation region related to $\sigma_{ESS} = 30$. This insight suggests that an increase in the penalty coefficient σ_{ESS} , if appropriately adjusted, can guarantee the exactness of the relaxation, validating the effectiveness of the given practical relaxation method presented in Subsection B of Section IV.

5.6. Demonstrating the influence of increasing the penalty parameter σ_{ESS} on the charging and discharging results of energy storage

Given that $a_t^{dc} = 15$ \$/MWh, $a_t^{ch} = -10$ \$/MWh and the renewable generation cost is -5 \$/MWh, we then demonstrate the influence of increasing the penalty parameter σ_{ESS} on the charging and discharging results of energy storage. Here, the results corresponding to $\sigma_{ESS} = 3$ are used as the benchmark. The relative differences in charging and discharging results under different penalty parameter σ_{ESS} can be calculated by (33).

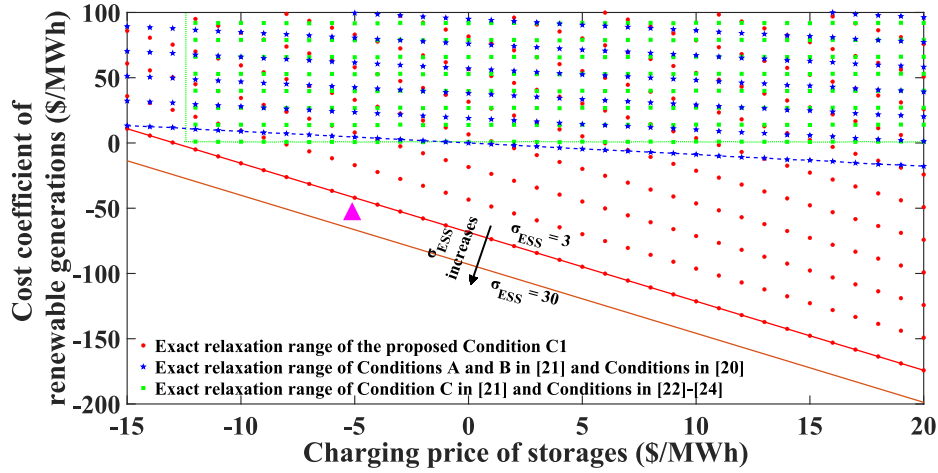


Fig. 11. The exact relaxation range for storages under varying charging price and renewable energy cost coefficient.

$$\max \left(\frac{|p_{ESS,n,t}^{ch,\sigma} - p_{ESS,n,t}^{ch,\sigma=3}|}{p_{ESS,n,t}^{ch,\sigma=3}}, \frac{|p_{ESS,n,t}^{dc,\sigma} - p_{ESS,n,t}^{dc,\sigma=3}|}{p_{ESS,n,t}^{dc,\sigma=3}} \right), \forall t \in \mathcal{T}, n \in \mathcal{S}_{ESS} \quad (33)$$

where $p_{ESS,n,t}^{ch,\sigma=3}$ and $p_{ESS,n,t}^{dc,\sigma=3}$ are the charging and discharging active power of the n^{th} ESS at time t when $\sigma_{ESS} = 3$. $p_{ESS,n,t}^{ch,\sigma}$ and $p_{ESS,n,t}^{dc,\sigma}$ are the charging and discharging active power of the n^{th} ESS at time t when σ_{ESS} is assigned the value of σ .

We can observe that although the value of σ_{ESS} has already been increased by a hundred times, the relative differences in charging and discharging results are minimal, as illustrated by Fig. 12. Furthermore, taking $\sigma_{ESS} = 3$ and $\sigma_{ESS} = 30$ as examples, Fig. 13 presents their charging and discharging results of storages. The marginal differences can usually be neglected in practical operation scenarios, which also highlights the broad application prospects of the proposed practical relaxation method.

Furthermore, extensive numerical tests have been performed over different penalty parameter values, multiple system scales, and charging/discharging prices to verify the applicability of the proposed relaxation condition. Due to space limitations, refer to Appendix H in the supplementary file [31] for details.

6. Conclusion

In this paper, an exact GRC with a priori property is first proposed. The developed GRC provably ensure no SCD in the relaxed power grid

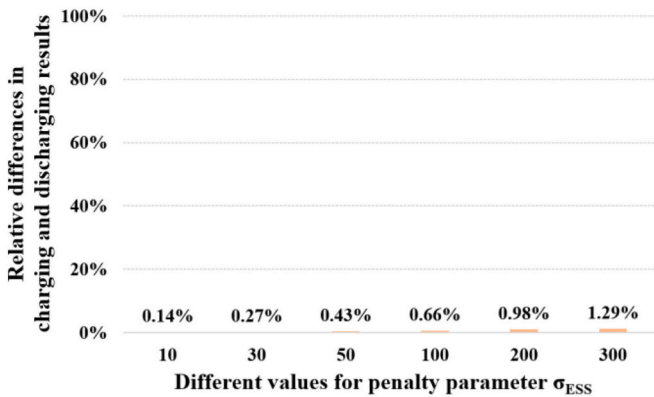
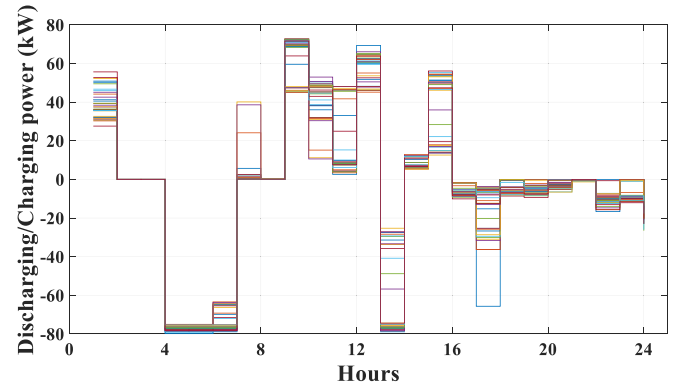
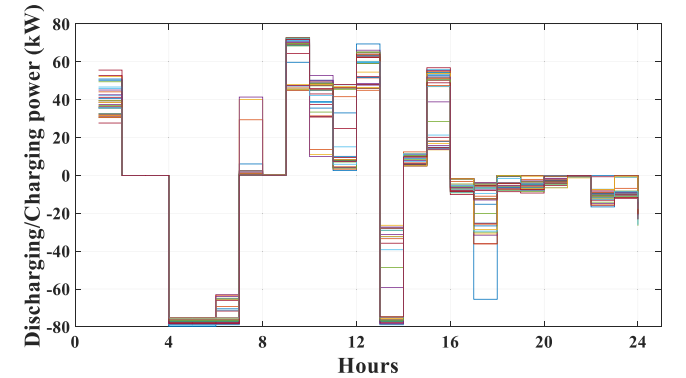


Fig. 12. Illustration of the relative differences in charging and discharging results under different penalty parameter σ_{ESS} , compared with those of $\sigma_{ESS} = 3$.



(a) $\sigma_{ESS} = 3$



(b) $\sigma_{ESS} = 30$

Fig. 13. Illustration of the charging and discharging results of storages under different penalty parameter σ_{ESS} .

optimization problems, excluding the complementarity constraints related to ESSs. We also prove that the exact relaxation region on LMPs formed by the proposed GRC contains the others. Then, by introducing penalties for charging and discharging losses of storage into the objective function, the applicability of the proposed GRC can be extended to the scenarios with arbitrary electricity prices. Moreover, we propose a criterion for determining the penalty parameter that ensures the compliance of the proposed GRC. In addition, in combination with a priori property of the proposed GRC, a practical relaxation method is presented. Finally, we also propose a practical two-stage penalty

parameter adjustment scheme to weaken the dependence of the proposed relaxation method on the prediction accuracy of the LMPs.

The exactness and advantages of the proposed GRC are comprehensively demonstrated in numerical tests:

- i) Extensive numerical tests under various positive/negative electricity price scenarios have demonstrated the broad applicability of the proposed GRC.
- ii) Even during periods of negative net load, the proposed GRC can still ensure the exactness of the relaxation.
- iii) We further characterize the exact relaxation ranges of different relaxation methods with multiple penalty parameter values. The results highlight the remarkable superiority of the proposed relaxation method over existing relaxation conditions, and confirm its property of not being significantly dependent on the prediction accuracy of the LMPs.

In summary, the numerical results show the broad application prospects of the proposed relaxation method in the operations of power systems with a high penetration of renewable energy and ESSs.

It is worth mentioning that the duality theory can be further employed to perform in-depth theoretical analysis based on the proposed exact GRC. Accordingly, it facilitates the optimization of ESSs to be embedded in more complicated problems, such as multi-level optimization problems [40] and robust optimization problems [41].

In future research, we intend to further extend the proposed relaxation method to enhance its generality, rendering it widely applicable to a class of optimization problems with complementary constraints.

Appendix A. Proof for condition C1

Proof of Condition C1: Denoting the Lagrangian function of the relaxed model as L_{RM} , and defining a new variable $\Gamma_{ESS,n,t} = \sum_{\tau=t}^T (1 - \delta_{ESS,n,\tau})^{\tau-t} (\lambda_{ESS,n,\tau}^{SOC,2} - \lambda_{ESS,n,\tau}^{SOC,1})$. The constraints related to $p_{ESS,n,t}^{ch}$ and $p_{ESS,n,t}^{dc}$ include (4), (6)–(11), (13)–(14), (16). Then employing KKT conditions, the following equations are obtained:

$$\frac{\partial L_{RM}}{\partial (p_{ESS,n,t}^{ch})} = \frac{\partial obj}{\partial (p_{ESS,n,t}^{ch})} + \lambda_{j,t}^p - \lambda_{ESS,n,t}^{ch,1} + \lambda_{ESS,n,t}^{ch,2} + \frac{\lambda_{ESS,n,t}^{relax}}{\bar{P}_{ESS,n}^{ch}} + 2\lambda_{ESS,n,t}^{S,2} \cdot p_{ESS,n,t}^{ch} + \eta_{ESS,n}^{ch} \cdot \Gamma_{ESS,n,t} \cdot \Delta t + \lambda_{ESS,n,t}^{RD,1} = 0 \quad (34)$$

$$\frac{\partial L_{RM}}{\partial (p_{ESS,n,t}^{dc})} = \frac{\partial obj}{\partial (p_{ESS,n,t}^{dc})} - \lambda_{j,t}^p - \lambda_{ESS,n,t}^{dc,1} + \lambda_{ESS,n,t}^{dc,2} + \frac{\lambda_{ESS,n,t}^{relax}}{\bar{P}_{ESS,n}^{dc}} + 2\lambda_{ESS,n,t}^{S,1} \cdot p_{ESS,n,t}^{dc} - \Gamma_{ESS,n,t} \cdot \Delta t / \eta_{ESS,n}^{dc} + \lambda_{ESS,n,t}^{RU,1} = 0 \quad (35)$$

With (34)/ $\eta_{ESS,n}^{dc}$ + (35) $\cdot \eta_{ESS,n}^{ch}$, we have

$$\left[\frac{\partial obj}{\partial (p_{ESS,n,t}^{ch})} + \lambda_{j,t}^p - \lambda_{ESS,n,t}^{ch,1} + \lambda_{ESS,n,t}^{ch,2} + 2\lambda_{ESS,n,t}^{S,2} \cdot p_{ESS,n,t}^{ch} + \frac{\lambda_{ESS,n,t}^{relax}}{\bar{P}_{ESS,n}^{ch}} + \lambda_{ESS,n,t}^{RD,1} \right] / \eta_{ESS,n}^{dc} + \left[\frac{\partial obj}{\partial (p_{ESS,n,t}^{dc})} - \lambda_{j,t}^p - \lambda_{ESS,n,t}^{dc,1} + \lambda_{ESS,n,t}^{dc,2} + \frac{\lambda_{ESS,n,t}^{relax}}{\bar{P}_{ESS,n}^{dc}} + 2\lambda_{ESS,n,t}^{S,1} \cdot p_{ESS,n,t}^{dc} + \lambda_{ESS,n,t}^{RU,1} \right] \cdot \eta_{ESS,n}^{ch} = 0 \quad (36)$$

Further, eq. (36) yields:

CRediT authorship contribution statement

Qi Wang: Writing – original draft, Visualization, Validation, Software, Methodology. **Wenchuan Wu:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization. **Chenhui Lin:** Writing – review & editing, Conceptualization. **Shuwei Xu:** Writing – review & editing, Conceptualization. **Siyuan Wang:** Conceptualization, Writing – review & editing. **Jian Tian:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

We have shared a link to the data in the reference.

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$$\begin{aligned}
& \lambda_{j,t}^p - \left[\frac{\partial \text{obj}}{\partial (p_{\text{ESS},n,t}^{\text{ch}})} \middle/ \eta_{\text{ESS},n}^{\text{dc}} + \frac{\partial \text{obj}}{\partial (p_{\text{ESS},n,t}^{\text{dc}})} \eta_{\text{ESS},n}^{\text{ch}} \right] \middle/ \left(\eta_{\text{ESS},n}^{\text{ch}} - \frac{1}{\eta_{\text{ESS},n}^{\text{dc}}} \right) \\
&= \left(\lambda_{\text{ESS},n,t}^{\text{ch},1} \middle/ \eta_{\text{ESS},n}^{\text{dc}} + \lambda_{\text{ESS},n,t}^{\text{dc},1} \eta_{\text{ESS},n}^{\text{ch}} \right) \middle/ \left(1 \middle/ \eta_{\text{ESS},n}^{\text{dc}} - \eta_{\text{ESS},n}^{\text{ch}} \right) \\
&\quad - \left\{ \left[\lambda_{\text{ESS},n,t}^{\text{ch},2} + 2\lambda_{\text{ESS},n,t}^{\text{S},2} p_{\text{ESS},n,t}^{\text{ch}} + \frac{\lambda_{\text{ESS},n,t}^{\text{relax}}}{\bar{p}_{\text{ESS},n}^{\text{ch}}} + \lambda_{\text{ESS},n,t}^{\text{RD},1} \right] \middle/ \eta_{\text{ESS},n}^{\text{dc}} \right. \\
&\quad \left. + \left[\lambda_{\text{ESS},n,t}^{\text{dc},2} + 2\lambda_{\text{ESS},n,t}^{\text{S},1} p_{\text{ESS},n,t}^{\text{dc}} + \frac{\lambda_{\text{ESS},n,t}^{\text{relax}}}{\bar{p}_{\text{ESS},n}^{\text{dc}}} + \lambda_{\text{ESS},n,t}^{\text{RU},1} \right] \eta_{\text{ESS},n}^{\text{ch}} \right\} \middle/ \left(\frac{1}{\eta_{\text{ESS},n}^{\text{dc}}} - \eta_{\text{ESS},n}^{\text{ch}} \right)
\end{aligned} \tag{37}$$

In combination with the expression of $\lambda_{j,t}^p$ in eq. (37), we can conclude that if the proposed *Condition C1* holds, we can obtain (38).

$$\begin{aligned}
& \left(\lambda_{\text{ESS},n,t}^{\text{ch},1} \middle/ \eta_{\text{ESS},n}^{\text{dc}} + \lambda_{\text{ESS},n,t}^{\text{dc},1} \eta_{\text{ESS},n}^{\text{ch}} \right) \middle/ \left(1 \middle/ \eta_{\text{ESS},n}^{\text{dc}} - \eta_{\text{ESS},n}^{\text{ch}} \right) > \\
& \left\{ \left[\lambda_{\text{ESS},n,t}^{\text{ch},2} + 2\lambda_{\text{ESS},n,t}^{\text{S},2} p_{\text{ESS},n,t}^{\text{ch}} + \frac{\lambda_{\text{ESS},n,t}^{\text{relax}}}{\bar{p}_{\text{ESS},n}^{\text{ch}}} + \lambda_{\text{ESS},n,t}^{\text{RD},1} \right] \middle/ \eta_{\text{ESS},n}^{\text{dc}} \right. \\
& \left. + \left[\lambda_{\text{ESS},n,t}^{\text{dc},2} + 2\lambda_{\text{ESS},n,t}^{\text{S},1} p_{\text{ESS},n,t}^{\text{dc}} + \frac{\lambda_{\text{ESS},n,t}^{\text{relax}}}{\bar{p}_{\text{ESS},n}^{\text{dc}}} + \lambda_{\text{ESS},n,t}^{\text{RU},1} \right] \eta_{\text{ESS},n}^{\text{ch}} \right\} \middle/ \left(1 \middle/ \eta_{\text{ESS},n}^{\text{dc}} - \eta_{\text{ESS},n}^{\text{ch}} \right)
\end{aligned} \tag{38}$$

Considering $\eta_{\text{ESS},n}^{\text{dc}}, \eta_{\text{ESS},n}^{\text{ch}} > 0$, $\lambda_{\text{ESS},n,t}^{\text{ch},2}, \lambda_{\text{ESS},n,t}^{\text{dc},2} \geq 0$ and $\lambda_{\text{ESS},n,t}^{\text{S},1}, \lambda_{\text{ESS},n,t}^{\text{S},2}, \lambda_{\text{ESS},n,t}^{\text{relax}}, \lambda_{\text{ESS},n,t}^{\text{RD},1}, \lambda_{\text{ESS},n,t}^{\text{RU},1}, p_{\text{ESS},n,t}^{\text{ch}}, p_{\text{ESS},n,t}^{\text{dc}} \geq 0$, we can derive

$$\begin{aligned}
& \left[\lambda_{\text{ESS},n,t}^{\text{ch},2} + 2\lambda_{\text{ESS},n,t}^{\text{S},2} p_{\text{ESS},n,t}^{\text{ch}} + \frac{\lambda_{\text{ESS},n,t}^{\text{relax}}}{\bar{p}_{\text{ESS},n}^{\text{ch}}} + \lambda_{\text{ESS},n,t}^{\text{RD},1} \right] \middle/ \eta_{\text{ESS},n}^{\text{dc}} \\
& + \left[\lambda_{\text{ESS},n,t}^{\text{dc},2} + 2\lambda_{\text{ESS},n,t}^{\text{S},1} p_{\text{ESS},n,t}^{\text{dc}} + \frac{\lambda_{\text{ESS},n,t}^{\text{relax}}}{\bar{p}_{\text{ESS},n}^{\text{dc}}} + \lambda_{\text{ESS},n,t}^{\text{RU},1} \right] \eta_{\text{ESS},n}^{\text{ch}} \geq 0
\end{aligned} \tag{39}$$

Combining formulas (38), (39) and $1/\eta_{\text{ESS},n}^{\text{dc}} > \eta_{\text{ESS},n}^{\text{ch}}$, we have

$$\left(\lambda_{\text{ESS},n,t}^{\text{ch},1} \middle/ \eta_{\text{ESS},n}^{\text{dc}} + \lambda_{\text{ESS},n,t}^{\text{dc},1} \eta_{\text{ESS},n}^{\text{ch}} \right) > 0 \tag{40}$$

Since $\eta_{\text{ESS},n}^{\text{dc}}, \eta_{\text{ESS},n}^{\text{ch}} > 0$ and $\lambda_{\text{ESS},n,t}^{\text{ch},1}, \lambda_{\text{ESS},n,t}^{\text{dc},1} \geq 0$, (41) holds.

$$\lambda_{\text{ESS},n,t}^{\text{ch},1} + \lambda_{\text{ESS},n,t}^{\text{dc},1} > 0 \tag{41}$$

Subsequently, in combination with the complementary slackness condition, it follows that

$$\begin{aligned}
& \lambda_{\text{ESS},n,t}^{\text{ch},1} p_{\text{ESS},n,t}^{\text{ch}} = 0 \\
& \lambda_{\text{ESS},n,t}^{\text{dc},1} p_{\text{ESS},n,t}^{\text{dc}} = 0
\end{aligned} \tag{42}$$

Therefore, *Condition C1* indicates that $p_{\text{ESS},n,t}^{\text{ch}} p_{\text{ESS},n,t}^{\text{dc}} = 0$. In other words, based on *Condition C1*, the optimal solution enforces the complementary constraint, even though it is relaxed. ■

A.

Proof for the relaxation conditions of group B in [21] when adopting the established optimization model

The **Relaxation conditions of group B** in [21] are given as following. Although they are obtained using the ED model, similar relaxation conditions can be derived for the established optimization model.

$$\text{C3.1} : \frac{\partial \text{obj}}{\partial (p_{\text{ESS},n,t}^{\text{ch}})} + \frac{\partial \text{obj}}{\partial (p_{\text{ESS},n,t}^{\text{dc}})} > 0 \tag{43}$$

$$\text{C3.2} : \lambda_{j,t}^p \geq \lambda_{j,t}^{\text{p,C3}} = - \frac{\partial \text{obj}}{\partial (p_{\text{ESS},n,t}^{\text{ch}})} \tag{44}$$

Proof: We prove this type of conditions by contradiction. Suppose there exists $p_{\text{ESS},n,t}^{\text{ch}} > 0$ and $p_{\text{ESS},n,t}^{\text{dc}} > 0$ for the n^{th} ESS at time t in the optimal solution of the relaxed model. Then we have $\lambda_{\text{ESS},n,t}^{\text{ch},1} = 0$ and $\lambda_{\text{ESS},n,t}^{\text{dc},1} = 0$.

i) Combined with $\lambda_{\text{ESS},n,t}^{\text{ch},1} = 0$ and $\lambda_{\text{ESS},n,t}^{\text{ch},2} \geq 0$, formula (34) yields:

$$\frac{\partial \text{obj}}{\partial (p_{\text{ESS},n,t}^{\text{ch}})} + \lambda_{j,t}^p + \frac{\lambda_{\text{ESS},n,t}^{\text{relax}}}{\bar{p}_{\text{ESS},n}^{\text{ch}}} + 2\lambda_{\text{ESS},n,t}^{\text{S},2} p_{\text{ESS},n,t}^{\text{ch}} + \eta_{\text{ESS},n}^{\text{ch}} \Gamma_{\text{ESS},n,t} \Delta t + \lambda_{\text{ESS},n,t}^{\text{RD},1} \leq 0 \tag{45}$$

Furthermore, with $\lambda_{\text{ESS},n,t}^{\text{relax}}, \lambda_{\text{ESS},n,t}^{\text{S},2}, \lambda_{\text{ESS},n,t}^{\text{RD},1} \geq 0$, $\eta_{\text{ESS},n}^{\text{ch}} > 0$ and **condition C3.2**, we can infer that $\Gamma_{\text{ESS},n,t} \leq 0$.

ii) Summing formulas (34) and (35), and combining $\lambda_{\text{ESS},n,t}^{\text{ch},1} = \lambda_{\text{ESS},n,t}^{\text{dc},1} = 0$, we can obtain

$$\begin{aligned}
& \frac{\partial \text{obj}}{\partial (p_{\text{ESS},n,t}^{\text{ch}})} + \frac{\partial \text{obj}}{\partial (p_{\text{ESS},n,t}^{\text{dc}})} + \frac{\lambda_{\text{ESS},n,t}^{\text{relax}}}{\bar{P}_{\text{ESS},n}} + \frac{\lambda_{\text{ESS},n,t}^{\text{relax}}}{\bar{P}_{\text{ESS},n}} + \lambda_{\text{ESS},n,t}^{\text{ch},2} + \lambda_{\text{ESS},n,t}^{\text{dc},2} \\
& + \lambda_{\text{ESS},n,t}^{\text{RD},1} + \lambda_{\text{ESS},n,t}^{\text{RU},1} + 2 \left(\lambda_{\text{ESS},n,t}^{\text{S},2} \cdot p_{\text{ESS},n,t}^{\text{ch}} + \lambda_{\text{ESS},n,t}^{\text{S},1} \cdot p_{\text{ESS},n,t}^{\text{dc}} \right) \\
& - \left(1 / \eta_{\text{ESS},n}^{\text{dc}} - \eta_{\text{ESS},n}^{\text{ch}} \right) \cdot \Gamma_{\text{ESS},n,t} \cdot \Delta t = 0
\end{aligned} \tag{46}$$

In consideration of **condition C3_1**, $\lambda_{\text{ESS},n,t}^{\text{ch},2}, \lambda_{\text{ESS},n,t}^{\text{dc},2} \geq 0$, $\lambda_{\text{ESS},n,t}^{\text{relax}}, \lambda_{\text{ESS},n,t}^{\text{RD},1}, \lambda_{\text{ESS},n,t}^{\text{RU},1}, \lambda_{\text{ESS},n,t}^{\text{S},1}, \lambda_{\text{ESS},n,t}^{\text{S},2} \geq 0$, and $(1 / \eta_{\text{ESS},n}^{\text{dc}} - \eta_{\text{ESS},n}^{\text{ch}}) > 0$, it can be concluded that $\Gamma_{\text{ESS},n,t} > 0$, which conflicts with the previous conclusion $\Gamma_{\text{ESS},n,t} \leq 0$.

Therefore, we can draw the conclusion that $p_{\text{ESS},n,t}^{\text{ch}} > 0$ and $p_{\text{ESS},n,t}^{\text{dc}} > 0$ are impossible to appear in the optimal solution of the relaxed model at the same time for any ESS at any time period on the premise that **condition C3** holds. ■

B. Proof for the Relaxation Conditions in [20] and Relaxation Conditions of Group A in [21] when adopting the established optimization model

Similarly, the **Relaxation conditions** in [20] and **Relaxation conditions of group A** in [21] also hold for the established optimization model. The difference between **condition C4** and **condition C3** lies in their different signs.

$$\text{C4_1} : \frac{\partial \text{obj}}{\partial (p_{\text{ESS},n,t}^{\text{ch}})} + \frac{\partial \text{obj}}{\partial (p_{\text{ESS},n,t}^{\text{dc}})} \geq 0 \tag{47}$$

$$\text{C4_2} : \lambda_{j,t}^{\text{p}} > \lambda_{j,t}^{\text{p,C4}} = - \frac{\partial \text{obj}}{\partial (p_{\text{ESS},n,t}^{\text{ch}})} \tag{48}$$

Proof: We prove this type of conditions by contradiction.

i) With **formula (45)**, $\lambda_{\text{ESS},n,t}^{\text{relax}}, \lambda_{\text{ESS},n,t}^{\text{S},2}, \lambda_{\text{ESS},n,t}^{\text{RD},1} \geq 0$, $\eta_{\text{ESS},n}^{\text{ch}} > 0$ and **condition C4_2**, we can infer that $\Gamma_{\text{ESS},n,t} < 0$.

ii) Combining **formula (46)**, **condition C4_1**, $\lambda_{\text{ESS},n,t}^{\text{ch},2}, \lambda_{\text{ESS},n,t}^{\text{dc},2} \geq 0$, $\lambda_{\text{ESS},n,t}^{\text{relax}}, \lambda_{\text{ESS},n,t}^{\text{RD},1}, \lambda_{\text{ESS},n,t}^{\text{RU},1}, \lambda_{\text{ESS},n,t}^{\text{S},1}, \lambda_{\text{ESS},n,t}^{\text{S},2} \geq 0$ and $(1 / \eta_{\text{ESS},n}^{\text{dc}} - \eta_{\text{ESS},n}^{\text{ch}}) > 0$, we can know that $\Gamma_{\text{ESS},n,t} \geq 0$, which is in conflict with the above conclusion $\Gamma_{\text{ESS},n,t} < 0$.

Thus, the relaxation has been proven to be exact for the given **condition C4**. ■

C. Proof for the Relaxation Conditions of Group C in [21] when adopting the established optimization model

Under the **Relaxation conditions of Group C** in [21], the ESSs' complementarity constraints are enforced in ED. Here, we give the proof of these conditions in scenarios with the established optimization model:

$$\text{C5_1} : \frac{\partial \text{obj}}{\partial (p_{\text{ESS},n,t}^{\text{ch}})} / \eta_{\text{ESS},n}^{\text{dc}} + \frac{\partial \text{obj}}{\partial (p_{\text{ESS},n,t}^{\text{dc}})} \cdot \eta_{\text{ESS},n}^{\text{ch}} > 0 \tag{49}$$

$$\text{C5_2} : \lambda_{j,t}^{\text{p}} \geq 0 \tag{50}$$

Proof: We prove this type of conditions by contradiction. Assume $p_{\text{ESS},n,t}^{\text{ch}} \cdot p_{\text{ESS},n,t}^{\text{dc}} > 0$, then $\lambda_{\text{ESS},n,t}^{\text{ch},1} = \lambda_{\text{ESS},n,t}^{\text{dc},1} = 0$. With (34) + (35) · $\eta_{\text{ESS},n}^{\text{ch}} \cdot \eta_{\text{ESS},n}^{\text{dc}}$, we have

$$\begin{aligned}
& \left[\frac{\partial \text{obj}}{\partial (p_{\text{ESS},n,t}^{\text{ch}})} + \lambda_{j,t}^{\text{p}} - \lambda_{\text{ESS},n,t}^{\text{ch},1} + \lambda_{\text{ESS},n,t}^{\text{ch},2} + 2\lambda_{\text{ESS},n,t}^{\text{S},2} \cdot p_{\text{ESS},n,t}^{\text{ch}} \right. \\
& \left. + \frac{\lambda_{\text{ESS},n,t}^{\text{relax}}}{\bar{P}_{\text{ESS},n}} + \lambda_{\text{ESS},n,t}^{\text{RD},1} \right] + \left[\frac{\partial \text{obj}}{\partial (p_{\text{ESS},n,t}^{\text{dc}})} - \lambda_{j,t}^{\text{p}} - \lambda_{\text{ESS},n,t}^{\text{dc},1} + \lambda_{\text{ESS},n,t}^{\text{dc},2} \right. \\
& \left. + 2\lambda_{\text{ESS},n,t}^{\text{S},1} \cdot p_{\text{ESS},n,t}^{\text{dc}} + \frac{\lambda_{\text{ESS},n,t}^{\text{relax}}}{\bar{P}_{\text{ESS},n}} + \lambda_{\text{ESS},n,t}^{\text{RU},1} \right] \cdot \eta_{\text{ESS},n}^{\text{ch}} \cdot \eta_{\text{ESS},n}^{\text{dc}} = 0
\end{aligned} \tag{51}$$

Further, eq. (51) yields:

$$\begin{aligned}
& \lambda_{j,t}^{\text{p}} (1 - \eta_{\text{ESS},n}^{\text{ch}} \cdot \eta_{\text{ESS},n}^{\text{dc}}) + \frac{\partial \text{obj}}{\partial (p_{\text{ESS},n,t}^{\text{ch}})} + \frac{\partial \text{obj}}{\partial (p_{\text{ESS},n,t}^{\text{dc}})} \cdot \eta_{\text{ESS},n}^{\text{ch}} \cdot \eta_{\text{ESS},n}^{\text{dc}} \\
& + \lambda_{\text{ESS},n,t}^{\text{ch},2} + 2\lambda_{\text{ESS},n,t}^{\text{S},2} \cdot p_{\text{ESS},n,t}^{\text{ch}} + \frac{\lambda_{\text{ESS},n,t}^{\text{relax}}}{\bar{P}_{\text{ESS},n}} + \lambda_{\text{ESS},n,t}^{\text{RD},1} \\
& + \left(\lambda_{\text{ESS},n,t}^{\text{dc},2} + 2\lambda_{\text{ESS},n,t}^{\text{S},1} \cdot p_{\text{ESS},n,t}^{\text{dc}} + \frac{\lambda_{\text{ESS},n,t}^{\text{relax}}}{\bar{P}_{\text{ESS},n}} + \lambda_{\text{ESS},n,t}^{\text{RU},1} \right) \cdot \eta_{\text{ESS},n}^{\text{ch}} \cdot \eta_{\text{ESS},n}^{\text{dc}} = 0
\end{aligned} \tag{52}$$

Given **condition C5** and $\eta_{\text{ESS},n}^{\text{ch}} \cdot \eta_{\text{ESS},n}^{\text{dc}} < 1$, it follows that

$$\lambda_{j,t}^p \left(1 - \eta_{ESS,n}^{\text{ch}} \cdot \eta_{ESS,n}^{\text{dc}}\right) + \frac{\partial \text{obj}}{\partial (p_{ESS,n,t}^{\text{ch}})} + \frac{\partial \text{obj}}{\partial (p_{ESS,n,t}^{\text{dc}})} \cdot \eta_{ESS,n}^{\text{ch}} \cdot \eta_{ESS,n}^{\text{dc}} > 0 \quad (53)$$

Further, combining formulas (52) and (53), we have

$$\begin{aligned} & \lambda_{ESS,n,t}^{\text{ch},2} + 2\lambda_{ESS,n,t}^{\text{S},2} \cdot p_{ESS,n,t}^{\text{ch}} + \frac{\lambda_{ESS,n,t}^{\text{relax}}}{P_{ESS,n}^{\text{ch}}} + \lambda_{ESS,n,t}^{\text{RD},1} \\ & + \left(\lambda_{ESS,n,t}^{\text{dc},2} + 2\lambda_{ESS,n,t}^{\text{S},1} \cdot p_{ESS,n,t}^{\text{dc}} + \frac{\lambda_{ESS,n,t}^{\text{relax}}}{P_{ESS,n}^{\text{dc}}} + \lambda_{ESS,n,t}^{\text{RU},1} \right) \cdot \eta_{ESS,n}^{\text{ch}} \cdot \eta_{ESS,n}^{\text{dc}} < 0 \end{aligned} \quad (54)$$

While as $\lambda_{ESS,n,t}^{\text{ch},2}, \lambda_{ESS,n,t}^{\text{dc},2}, \lambda_{ESS,n,t}^{\text{S},1}, \lambda_{ESS,n,t}^{\text{S},2}, \lambda_{ESS,n,t}^{\text{relax}}, \lambda_{ESS,n,t}^{\text{RD},1}, \lambda_{ESS,n,t}^{\text{RU},1} \geq 0$, there is a contradiction with formula (54). As a conclusion, under the given **condition C5**, the assumption $p_{ESS,n,t}^{\text{ch}}, p_{ESS,n,t}^{\text{dc}} > 0$ does not hold. Therefore, the relaxation conditions are exact. ■

D. Proof for the Relaxation Conditions in [22] when adopting the established optimization model

The **Relaxation conditions** in [22] are presented as follows:

$$\text{C6}_1: \frac{\partial \text{obj}}{\partial (p_{ESS,n,t}^{\text{ch}})} \geq 0, \frac{\partial \text{obj}}{\partial (p_{ESS,n,t}^{\text{dc}})} \geq 0, \frac{\partial \text{obj}}{\partial (p_{ESS,n,t}^{\text{ch}})} + \frac{\partial \text{obj}}{\partial (p_{ESS,n,t}^{\text{dc}})} > 0 \quad (55)$$

$$\text{C6}_2: \lambda_{j,t}^p \geq 0 \quad (56)$$

Proof: Comparing **Condition C5** and **Condition C6**, we can find that they differ only in condition C5_1 and condition C6_1. What's more, we can easily derive condition C5_1 from condition C6_1. Thus, we can conclude that the region formulated by **Condition C5** contains that of **Condition C6**. Then, combined with the proof in the previous subsection C in this appendix, we can draw the conclusion that the relaxation is exact for the given **Condition C6** when adopting the established optimization model. ■

Appendix C. Proof for Lemma 1

In addition to the relaxation conditions in [20–22] already presented in Appendix B, we also provide the relaxation conditions in [23,24] here to facilitate subsequent proofs.

In [24], the storage penalty term similar to (24) is added in the objective function to avoid SCD. And the following conditions are given:

$$\text{C7}_1: \lambda_{j,t}^p \geq 0 \quad (57)$$

$$\text{C7}_2: \lambda_{ESS,n,t}^{\text{S},1} = \lambda_{ESS,n,t}^{\text{S},2} = 0 \quad (58)$$

The relaxation conditions in [23] are presented as:

$$\text{C8}_1: \frac{\partial \text{obj}}{\partial (p_{ESS,n,t}^{\text{ch}})} = 0, \frac{\partial \text{obj}}{\partial (p_{ESS,n,t}^{\text{dc}})} > 0 \quad (59)$$

$$\text{C8}_2: \lambda_{j,t}^p > 0 \quad (60)$$

Subsequently, we redescribe Lemma 1 as Lemmas 4–9 equivalently. Correspondingly, if Lemmas 4–9 all hold, it can be concluded that Lemma 1 is valid. Below, we will provide the proofs for Lemmas 4–9. Specifically, Lemmas 4, 6–9 employ the direct proof methods, while proof by contradiction is used in Lemma 5.

A. Proof for Lemma 4

Lemma 4. The region about LMPs formulated by **Condition C1** contains that of the **Relaxation conditions of group B** in [21].

Proof: The **Condition C1** is

$$\lambda_{j,t}^p > \lambda_{j,t}^{\text{p,C1}} \quad (61)$$

And the **condition C3_2** in **Relaxation conditions of group B** in [21] is

$$\lambda_{j,t}^p \geq \lambda_{j,t}^{\text{p,C3}} \quad (62)$$

Then, we can conclude that if $\lambda_{j,t}^{\text{p,C3}} > \lambda_{j,t}^{\text{p,C1}}$, Lemma 4 holds. Below, we will provide the proof of $\lambda_{j,t}^{\text{p,C3}} > \lambda_{j,t}^{\text{p,C1}}$.

According to the **condition C3_1** and $0 < \eta_{ESS,n}^{\text{ch}} \cdot \eta_{ESS,n}^{\text{dc}} < 1$, we have

$$\frac{\partial \text{obj}}{\partial (p_{ESS,n,t}^{\text{ch}})} \cdot \eta_{ESS,n}^{\text{ch}} \cdot \eta_{ESS,n}^{\text{dc}} > - \frac{\partial \text{obj}}{\partial (p_{ESS,n,t}^{\text{dc}})} \cdot \eta_{ESS,n}^{\text{ch}} \cdot \eta_{ESS,n}^{\text{dc}} \quad (63)$$

Add term $-\frac{\partial \text{obj}}{\partial (p_{ESS,n,t}^{\text{ch}})}$ to both sides of formula (63), and it follows that

$$-\frac{\partial \text{obj}}{\partial (p_{\text{ESS},n,t}^{\text{ch}})} + \frac{\partial \text{obj}}{\partial (p_{\text{ESS},n,t}^{\text{ch}})} \eta_{\text{ESS},n}^{\text{ch}} \eta_{\text{ESS},n}^{\text{dc}} > -\frac{\partial \text{obj}}{\partial (p_{\text{ESS},n,t}^{\text{ch}})} - \frac{\partial \text{obj}}{\partial (p_{\text{ESS},n,t}^{\text{dc}})} \eta_{\text{ESS},n}^{\text{ch}} \eta_{\text{ESS},n}^{\text{dc}} \quad (64)$$

The above expression can be further expressed as:

$$-\frac{\partial \text{obj}}{\partial (p_{\text{ESS},n,t}^{\text{ch}})} (1 - \eta_{\text{ESS},n}^{\text{ch}} \eta_{\text{ESS},n}^{\text{dc}}) > -\frac{\partial \text{obj}}{\partial (p_{\text{ESS},n,t}^{\text{ch}})} - \frac{\partial \text{obj}}{\partial (p_{\text{ESS},n,t}^{\text{dc}})} \eta_{\text{ESS},n}^{\text{ch}} \eta_{\text{ESS},n}^{\text{dc}} \quad (65)$$

Further, divide both sides of inequality (64) by $(1 - \eta_{\text{ESS},n}^{\text{ch}} \eta_{\text{ESS},n}^{\text{dc}})$, we can obtain

$$\begin{aligned} \lambda_{j,t}^{\text{p,C3}} &= -\frac{\partial \text{obj}}{\partial (p_{\text{ESS},n,t}^{\text{ch}})} > \\ &\left[\frac{\partial \text{obj}}{\partial (p_{\text{ESS},n,t}^{\text{ch}})} \eta_{\text{ESS},n}^{\text{dc}} + \frac{\partial \text{obj}}{\partial (p_{\text{ESS},n,t}^{\text{dc}})} \eta_{\text{ESS},n}^{\text{ch}} \right] / \left(\eta_{\text{ESS},n}^{\text{ch}} - \frac{1}{\eta_{\text{ESS},n}^{\text{dc}}} \right) \\ &= \lambda_{j,t}^{\text{p,C1}} \end{aligned} \quad (66)$$

Thus, [Lemma 4](#) holds. ■

B. Proof for [Lemma 5](#)

Lemma 5. The region about LMPs formulated by **Condition C1** contains that of the **Relaxation conditions** in [\[24\]](#).

Proof: Recall that the condition given in **Condition C1** is

$$\lambda_{j,t}^{\text{p}} > \lambda_{j,t}^{\text{p,C1}} \quad (67)$$

Compared with **condition C7_1**, we can know that if $\lambda_{j,t}^{\text{p,C1}} < 0$, [Lemma 5](#) holds. Below, we will provide the proof of $\lambda_{j,t}^{\text{p,C1}} < 0$ on the premise that **condition C7** holds. We will give proofs by contradiction.

i) Assume $p_{\text{ESS},n,t}^{\text{ch}} p_{\text{ESS},n,t}^{\text{dc}} > 0$, then we have $\lambda_{\text{ESS},n,t}^{\text{ch},1} = \lambda_{\text{ESS},n,t}^{\text{dc},1} = 0$. Further, with (34), it follows that

$$\begin{aligned} \Gamma_{\text{ESS},n,t} &= -\frac{\partial \text{obj}}{\partial (p_{\text{ESS},n,t}^{\text{ch}})} \left/ \left(\eta_{\text{ESS},n}^{\text{ch}} \Delta t \right) - \left(\lambda_{j,t}^{\text{p}} - \lambda_{\text{ESS},n,t}^{\text{ch},1} \right) \right/ \left(\eta_{\text{ESS},n}^{\text{ch}} \Delta t \right) \\ &\quad - \left(\lambda_{\text{ESS},n,t}^{\text{ch},2} + \frac{\lambda_{\text{ESS},n,t}^{\text{relax}}}{P_{\text{ESS},n}^{\text{ch}}} + 2\lambda_{\text{ESS},n,t}^{\text{S},2} p_{\text{ESS},n,t}^{\text{ch}} + \lambda_{\text{ESS},n,t}^{\text{RD},1} \right) \left/ \left(\eta_{\text{ESS},n}^{\text{ch}} \Delta t \right) \right. \end{aligned} \quad (68)$$

Combining **condition C7** and $\lambda_{\text{ESS},n,t}^{\text{ch},1} = 0$, $\eta_{\text{ESS},n}^{\text{ch}} > 0$, $\lambda_{\text{ESS},n,t}^{\text{ch},2}$, $\lambda_{\text{ESS},n,t}^{\text{relax}}$, $\lambda_{\text{ESS},n,t}^{\text{S},2}$, $\lambda_{\text{ESS},n,t}^{\text{RD},1} \geq 0$, we can obtain [formula \(69\)](#) from (68).

$$\Gamma_{\text{ESS},n,t} \leq -\frac{\partial \text{obj}}{\partial (p_{\text{ESS},n,t}^{\text{ch}})} \left/ \left(\eta_{\text{ESS},n}^{\text{ch}} \Delta t \right) \right. \quad (69)$$

ii) Subsequently, we can derive [formula \(70\)](#) by summing [formulas \(34\) and \(35\)](#).

$$\begin{aligned} 0 &= \lambda_{\text{ESS},n,t}^{\text{ch},1} + \lambda_{\text{ESS},n,t}^{\text{dc},1} = \frac{\partial \text{obj}}{\partial (p_{\text{ESS},n,t}^{\text{ch}})} + \frac{\partial \text{obj}}{\partial (p_{\text{ESS},n,t}^{\text{dc}})} + \frac{\lambda_{\text{ESS},n,t}^{\text{relax}}}{P_{\text{ESS},n}^{\text{ch}}} + \frac{\lambda_{\text{ESS},n,t}^{\text{relax}}}{P_{\text{ESS},n}^{\text{dc}}} \\ &\quad + \lambda_{\text{ESS},n,t}^{\text{ch},2} + \lambda_{\text{ESS},n,t}^{\text{dc},2} + 2 \left(\lambda_{\text{ESS},n,t}^{\text{S},2} p_{\text{ESS},n,t}^{\text{ch}} + \lambda_{\text{ESS},n,t}^{\text{S},1} p_{\text{ESS},n,t}^{\text{dc}} \right) \\ &\quad + \lambda_{\text{ESS},n,t}^{\text{RD},1} + \lambda_{\text{ESS},n,t}^{\text{RU},1} - \left(1 / \eta_{\text{ESS},n}^{\text{dc}} - \eta_{\text{ESS},n}^{\text{ch}} \right) \cdot \Gamma_{\text{ESS},n,t} \Delta t \end{aligned} \quad (70)$$

Further, eq. (70) yields:

$$\begin{aligned} \Gamma_{\text{ESS},n,t} &= \left(\frac{\partial \text{obj}}{\partial (p_{\text{ESS},n,t}^{\text{ch}})} + \frac{\partial \text{obj}}{\partial (p_{\text{ESS},n,t}^{\text{dc}})} \right) \left/ \left[\left(\frac{1}{\eta_{\text{ESS},n}^{\text{dc}}} - \eta_{\text{ESS},n}^{\text{ch}} \right) \Delta t \right] \right. \\ &\quad + \left(\frac{\lambda_{\text{ESS},n,t}^{\text{relax}}}{P_{\text{ESS},n}^{\text{ch}}} + \frac{\lambda_{\text{ESS},n,t}^{\text{relax}}}{P_{\text{ESS},n}^{\text{dc}}} + \lambda_{\text{ESS},n,t}^{\text{ch},2} + \lambda_{\text{ESS},n,t}^{\text{dc},2} \right) \left/ \left[\left(\frac{1}{\eta_{\text{ESS},n}^{\text{dc}}} - \eta_{\text{ESS},n}^{\text{ch}} \right) \Delta t \right] \right. \\ &\quad + 2 \left(\lambda_{\text{ESS},n,t}^{\text{S},2} p_{\text{ESS},n,t}^{\text{ch}} + \lambda_{\text{ESS},n,t}^{\text{S},1} p_{\text{ESS},n,t}^{\text{dc}} \right) \left/ \left[\left(1 / \eta_{\text{ESS},n}^{\text{dc}} - \eta_{\text{ESS},n}^{\text{ch}} \right) \Delta t \right] \right. \\ &\quad + \left(\lambda_{\text{ESS},n,t}^{\text{RD},1} + \lambda_{\text{ESS},n,t}^{\text{RU},1} \right) \left/ \left[\left(1 / \eta_{\text{ESS},n}^{\text{dc}} - \eta_{\text{ESS},n}^{\text{ch}} \right) \Delta t \right] \right. \end{aligned} \quad (71)$$

Given $\lambda_{\text{ESS},n,t}^{\text{ch},2}$, $\lambda_{\text{ESS},n,t}^{\text{dc},2}$, $\lambda_{\text{ESS},n,t}^{\text{relax}}$, $\lambda_{\text{ESS},n,t}^{\text{S},1}$, $\lambda_{\text{ESS},n,t}^{\text{S},2}$, $p_{\text{ESS},n,t}^{\text{ch}}$, $p_{\text{ESS},n,t}^{\text{dc}} \geq 0$, $\lambda_{\text{ESS},n,t}^{\text{RD},1}$, $\lambda_{\text{ESS},n,t}^{\text{RU},1} \geq 0$ and $1 / \eta_{\text{ESS},n}^{\text{dc}} > \eta_{\text{ESS},n}^{\text{ch}}$, we can obtain inequality (72) based on (71).

$$\Gamma_{ESS,n,t} \geq \left(\frac{\partial \text{obj}}{\partial (p_{ESS,n,t}^{\text{ch}})} + \frac{\partial \text{obj}}{\partial (p_{ESS,n,t}^{\text{dc}})} \right) / \left[\left(1/\eta_{ESS,n}^{\text{dc}} - \eta_{ESS,n}^{\text{ch}} \right) \cdot \Delta t \right] \quad (72)$$

iii) When [formulas \(69\) and \(72\)](#) cannot be established simultaneously, there will be a contradiction, indicating that the assumption $p_{ESS,n,t}^{\text{ch}}, p_{ESS,n,t}^{\text{dc}} > 0$ is not valid. In this case, the relaxation is exact for the given [condition C7](#). Thus, we can derive [formula \(73\)](#), which is essentially the necessary condition for the **Relaxation conditions** in [\[24\]](#) to hold.

$$\left(\frac{\partial \text{obj}}{\partial (p_{ESS,n,t}^{\text{ch}})} + \frac{\partial \text{obj}}{\partial (p_{ESS,n,t}^{\text{dc}})} \right) / \left[\left(1/\eta_{ESS,n}^{\text{dc}} - \eta_{ESS,n}^{\text{ch}} \right) \cdot \Delta t \right] > - \frac{\partial \text{obj}}{\partial (p_{ESS,n,t}^{\text{ch}})} / \left(\eta_{ESS,n}^{\text{ch}} \cdot \Delta t \right) \quad (73)$$

Further, reformulating (73), we have

$$0 < \frac{\partial \text{obj}}{\partial (p_{ESS,n,t}^{\text{ch}})} / \eta_{ESS,n}^{\text{dc}} + \frac{\partial \text{obj}}{\partial (p_{ESS,n,t}^{\text{dc}})} \cdot \eta_{ESS,n}^{\text{ch}} \quad (74)$$

With (74) and $1/\eta_{ESS,n}^{\text{dc}} > \eta_{ESS,n}^{\text{ch}}$, we can obtain

$$0 > \left[\frac{\partial \text{obj}}{\partial (p_{ESS,n,t}^{\text{ch}})} / \eta_{ESS,n}^{\text{dc}} + \frac{\partial \text{obj}}{\partial (p_{ESS,n,t}^{\text{dc}})} \cdot \eta_{ESS,n}^{\text{ch}} \right] / \left(\eta_{ESS,n}^{\text{ch}} - \frac{1}{\eta_{ESS,n}^{\text{dc}}} \right) = \lambda_{j,t}^{\text{p,C1}} \quad (75)$$

Thus, [Lemma 5](#) holds. ■

Discussion 2: Comparing the [condition C5](#) and [condition C7](#), we can find that [condition C5_2](#) and [condition C7_1](#) are consistent. Moreover, we also notice that [formula \(74\)](#) is exactly the [condition C5_1](#). Then, combined with the aforementioned conclusion that [formula \(74\)](#) is the necessary condition for the **Relaxation conditions** in [\[24\]](#) to hold, we can infer that [condition C5_1](#) can be obtained when the **Relaxation conditions** in [\[24\]](#) hold. Therefore, it can be concluded that the [condition C5](#), i.e., **Relaxation conditions of Group C** in [\[21\]](#), can be derived based on the **Relaxation conditions** in [\[24\]](#).

C. Proof for [Lemma 6](#)

Lemma 6. The region about LMPs formulated by [Condition C1](#) contains those of the **Relaxation conditions** in [\[20\]](#) and the **Relaxation conditions** of group A in [\[21\]](#).

Proof: The sufficient condition provided in [Condition C1](#) is

$$\lambda_{j,t}^{\text{p}} > \lambda_{j,t}^{\text{p,C1}} \quad (76)$$

And the [condition C4_2](#) is

$$\lambda_{j,t}^{\text{p}} > \lambda_{j,t}^{\text{p,C4}} \quad (77)$$

Then, we can conclude that if $\lambda_{j,t}^{\text{p,C4}} \geq \lambda_{j,t}^{\text{p,C1}}$, [Lemma 6](#) holds.

Below, we will provide the proof of $\lambda_{j,t}^{\text{p,C4}} \geq \lambda_{j,t}^{\text{p,C1}}$.

According to the [condition C4_1](#) and $0 < \eta_{ESS,n}^{\text{ch}} \cdot \eta_{ESS,n}^{\text{dc}} < 1$, we have

$$\frac{\partial \text{obj}}{\partial (p_{ESS,n,t}^{\text{ch}})} \cdot \eta_{ESS,n}^{\text{ch}} \cdot \eta_{ESS,n}^{\text{dc}} \geq - \frac{\partial \text{obj}}{\partial (p_{ESS,n,t}^{\text{dc}})} \cdot \eta_{ESS,n}^{\text{ch}} \cdot \eta_{ESS,n}^{\text{dc}} \quad (78)$$

Add term $-\frac{\partial \text{obj}}{\partial (p_{ESS,n,t}^{\text{ch}})}$ to both sides of [formula \(78\)](#), it follows that

$$-\frac{\partial \text{obj}}{\partial (p_{ESS,n,t}^{\text{ch}})} + \frac{\partial \text{obj}}{\partial (p_{ESS,n,t}^{\text{ch}})} \cdot \eta_{ESS,n}^{\text{ch}} \cdot \eta_{ESS,n}^{\text{dc}} \geq -\frac{\partial \text{obj}}{\partial (p_{ESS,n,t}^{\text{ch}})} - \frac{\partial \text{obj}}{\partial (p_{ESS,n,t}^{\text{dc}})} \cdot \eta_{ESS,n}^{\text{ch}} \cdot \eta_{ESS,n}^{\text{dc}} \quad (79)$$

Further, divide both sides of inequality (79) by $(1 - \eta_{ESS,n}^{\text{ch}} \cdot \eta_{ESS,n}^{\text{dc}})$, we can obtains

$$\begin{aligned} \lambda_{j,t}^{\text{p,C4}} &= -\frac{\partial \text{obj}}{\partial (p_{ESS,n,t}^{\text{ch}})} \geq \\ &\left[\frac{\partial \text{obj}}{\partial (p_{ESS,n,t}^{\text{ch}})} / \eta_{ESS,n}^{\text{dc}} + \frac{\partial \text{obj}}{\partial (p_{ESS,n,t}^{\text{dc}})} \cdot \eta_{ESS,n}^{\text{ch}} \right] / \left(\eta_{ESS,n}^{\text{ch}} - \frac{1}{\eta_{ESS,n}^{\text{dc}}} \right) \\ &= \lambda_{j,t}^{\text{p,C1}} \end{aligned} \quad (80)$$

Thus, [Lemma 6](#) holds. ■

D. Proof for Lemma 7

Lemma 7. The region about LMPs formulated by **Condition C1** contains that of the **Relaxation conditions of group C** in [21].

Proof: **Condition C5_2** in **Relaxation conditions of group C** in [21] is

$$\lambda_{j,t}^p \geq 0 \quad (81)$$

And the sufficient condition provided in **Condition C1** is

$$\lambda_{j,t}^p > \lambda_{j,t}^{p,C1} \quad (82)$$

Obviously, if $0 > \lambda_{j,t}^{p,C1}$, **Lemma 7** holds.

Combining **condition C5_1** and $1/\eta_{ESS,n}^{dc} > \eta_{ESS,n}^{ch}$, we have

$$0 > \left[\frac{\partial \text{obj}}{\partial (p_{ESS,n,t}^{ch})} \right] / \eta_{ESS,n}^{dc} + \frac{\partial \text{obj}}{\partial (p_{ESS,n,t}^{dc})} \eta_{ESS,n}^{ch} \left/ \left(\eta_{ESS,n}^{ch} - \frac{1}{\eta_{ESS,n}^{dc}} \right) \right. = \lambda_{j,t}^{p,C1} \quad (83)$$

Therefore, **Lemma 7** holds. ■

E. Proof for Lemma 8

Lemma 8. The region about LMPs formulated by **Condition C1** contains that of the **Relaxation conditions** in [22].

Proof: Combined with the proof in subsection D in Appendix B, we can know that the regions formulated by **Relaxation conditions of group C** in [21] contains that of **Relaxation conditions** in [22]. Then, with the proof of **Lemma 7**, we can draw the conclusion that **Lemma 8** holds. ■

F. Proof for Lemma 9

Lemma 9. The region about LMPs formulated by **Condition C1** contains that of the **Relaxation conditions** in [23].

Proof: Comparing **Condition C6**, that is, **Relaxation conditions** in [22] and **Condition C8**, we can easily conclude that the region formulated by **Relaxation conditions** in [22] contains that of **Relaxation conditions** in [23]. Thus, combining the proof of **Lemma 8**, we can conclude that **Lemma 9** holds. ■

Appendix D. The multi-period active and reactive power coordinated optimization problem

Below, a multi-period active and reactive power joint coordination optimization problem is presented, which considers energy and reserve services simultaneously. It is crucial to emphasize that the energy storage operational scenarios described in Section II exhibit a wide range of diversity, posing no specific requirements for the objective function forms and operational constraints. The subsequent model is just one of the typical application scenarios.

G. The objective function

The objective function aims at seeking the optimal portfolio of active and reactive power to minimize the total production cost, which takes into account various components, including the generation cost, the reserve costs of conventional units and ESSs, penalties associated with renewable energy curtailment, energy storage arbitrage, and penalty for the charging and discharging losses of storages.

$$\begin{aligned} \min \text{obj} = & \sum_{t=1}^T \left\{ \underbrace{\sum_{n \in \Phi_{ESS}} [f_{n,t}(p_{ESS,n,t}^{ch}) + g_{n,t}(p_{ESS,n,t}^{dc})]}_{\text{energy storage arbitrage}} \right. \\ & + \underbrace{\sum_{g \in \Phi_G} C_{g,t}(p_{g,t}^G)}_{\text{generation cost}} + \underbrace{\sum_{n \in \Phi_{RG}} C_{n,t}(p_{n,t}^{RG})}_{\text{penalty for renewable energy curtailment}} + \underbrace{\sum_{n \in \Phi_{RG}} C_{n,t}^{Pen}(p_{n,t}^{RG})}_{\text{penalty for renewable energy curtailment}} \\ & + \underbrace{\sum_{n \in \Phi_{ESS}} C_{n,t}^R(R_{n,t}^{ch,ESS-}, R_{n,t}^{dc,ESS-}, R_{n,t}^{ch,ESS+}, R_{n,t}^{dc,ESS+})}_{\text{reserve cost of ESSs}} \\ & \left. + \underbrace{\sum_{g \in \Phi_G} C_{g,t}^R(R_{g,t}^{G+}, R_{g,t}^{G-})}_{\text{reserve cost of ESSs}} \right\} + \underbrace{C_{ESS}^{Pen}}_{\text{penalty for the charging and discharge losses}} \end{aligned} \quad (84)$$

where $f_{n,t}$ and $g_{n,t}$ are the charging and discharging fees of the n^{th} ESS at time t . $C_{g,t}$ and $C_{n,t}$ denote the operational costs of the g^{th} generator and the n^{th} renewable generation (RG) at time t . It's worth mentioning that the introduction of $C_{n,t}$ is for simulating the scenario of negative prices during periods of high renewable energy generation. In actual dispatch problems, $C_{n,t}$ can be excluded without compromising the exactness and applicability of the proposed exact relaxation condition. $C_{n,t}^{\text{Pen}}$ represents the penalty for renewable energy curtailment. $C_{g,t}^{\text{R}}$ and $C_{n,t}^{\text{R}}$ are the reserve costs of conventional units and storages. $C_{\text{ESS}}^{\text{Pen}}$ is the penalty term for the charging and discharging losses of storages [19].

$$f_{n,t}(p_{\text{ESS},n,t}^{\text{ch}}) = a_t^{\text{ch}} \cdot p_{\text{ESS},n,t}^{\text{ch}} \quad (85)$$

$$g_{n,t}(p_{\text{ESS},n,t}^{\text{dc}}) = a_t^{\text{dc}} \cdot p_{\text{ESS},n,t}^{\text{dc}}$$

$$C_{g,t}(p_{g,t}^{\text{G}}) = a_{2,g}(p_{g,t}^{\text{G}})^2 + a_{1,g}p_{g,t}^{\text{G}} + a_{0,g} \quad (86)$$

$$C_{n,t}(p_{n,t}^{\text{RG}}) = b_{n,t}p_{n,t}^{\text{RG}} \quad (87)$$

$$C_{n,t}^{\text{Pen}}(p_{n,t}^{\text{RG}}) = \frac{\sigma_{\text{RG}}(p_{n,t}^{\text{RG}} - \bar{p}_{n,t}^{\text{RG}})^2}{\bar{p}_{n,t}^{\text{RG}}} \quad (88)$$

$$C_{g,t}^{\text{R}}(R_{g,t}^{\text{G}+}, R_{g,t}^{\text{G}-}) = c_{g,t}^{\text{G}+} \cdot R_{g,t}^{\text{G}+} + c_{g,t}^{\text{G}-} \cdot R_{g,t}^{\text{G}-} \quad (89)$$

$$\begin{aligned} C_{n,t}^{\text{R}}(R_{n,t}^{\text{ch,ESS-}}, R_{n,t}^{\text{dc,ESS-}}, R_{n,t}^{\text{ch,ESS+}}, R_{n,t}^{\text{dc,ESS+}}) \\ = c_{n,t}^{\text{ESS+}} \cdot (R_{n,t}^{\text{ch,ESS+}} + R_{n,t}^{\text{dc,ESS+}}) + c_{n,t}^{\text{ESS-}} \cdot (R_{n,t}^{\text{ch,ESS-}} + R_{n,t}^{\text{dc,ESS-}}) \end{aligned} \quad (90)$$

$$C_{\text{ESS}}^{\text{Pen}} = \sigma_{\text{ESS}} \sum_{t=1}^T \sum_{n \in \Phi_{\text{ESS}}} \left\{ p_{\text{ESS},n,t}^{\text{dc}} \left(\frac{1}{\eta_{\text{ESS},n}^{\text{dc}}} - 1 \right) + p_{\text{ESS},n,t}^{\text{ch}} \left(1 - \eta_{\text{ESS},n}^{\text{ch}} \right) \right\} \quad (91)$$

where $p_{\text{ESS},n,t}^{\text{dc}}$ and $p_{\text{ESS},n,t}^{\text{ch}}$ are the discharging and charging active power of the n^{th} ESS at time t . $p_{g,t}^{\text{G}}$ and $p_{n,t}^{\text{RG}}$ are the active power outputs of the g^{th} generator and the n^{th} RG at time t . $\bar{p}_{n,t}^{\text{RG}}$ is the predicted active power of the n^{th} RG at time t . $R_{g,t}^{\text{G}+}$ and $R_{g,t}^{\text{G}-}$ denote the upward/downward reserve contributions of the g^{th} generator at time t . $R_{n,t}^{\text{ch,ESS-}}, R_{n,t}^{\text{dc,ESS-}}, R_{n,t}^{\text{ch,ESS+}}, R_{n,t}^{\text{dc,ESS+}}$ represent the downward/upward reserves of the n^{th} storage when it is in the charging/discharging status at time t , respectively. a_t^{ch} and a_t^{dc} are the charging and discharging prices at time t . $a_{2,g}$, $a_{1,g}$ and $a_{0,g}$ are the coefficients of the quadratic, linear and constant terms in the operational cost of the g^{th} generator. $b_{n,t}$ is the cost coefficient of the n^{th} RG at time t , which can be set as the clearing price for renewable energy in the electricity market. $c_{g,t}^{\text{G}+}$ and $c_{g,t}^{\text{G}-}$ denote the upward/downward reserve price of the g^{th} generator at time t . $c_{n,t}^{\text{ESS+}}$ and $c_{n,t}^{\text{ESS-}}$ represent the upward/downward reserve price of the n^{th} storage at time t . $\eta_{\text{ESS},n}^{\text{ch}}$ and $\eta_{\text{ESS},n}^{\text{dc}}$ are the charging and discharging efficiency of the n^{th} ESS. T is the whole time period. σ_{RG} and σ_{ESS} are the penalty coefficients. Φ_{G} , Φ_{RG} and Φ_{ESS} are the index sets of generators, RGs and ESSs.

H. Operational constraints

Details about the nodal power balance constraint related to LMP and these operational constraints of ESSs can be found in (4)–(5) and (6)–(16), respectively. The other constraints, including power flow model, operational constraints of generators, renewable generations and Static VAR compensators (SVCs), limits on voltage magnitudes, line transmission capacities and the minimum reserve requirement, are elaborated as following:

1) Power flow model

For the sake of generalizability, the detailed form of the OPF model is not presented here. For instance, the power flow models in [26–29] can be adopted based on the requirements. Subsequently, we give the nodal net power injection equations:

$$p_{j,t} = \sum_{g \in \Phi_{\text{G},j}} p_{g,t}^{\text{G}} + \sum_{n \in \Phi_{\text{RG},j}} p_{n,t}^{\text{RG}} + \sum_{n \in \Phi_{\text{ESS},j}} (p_{\text{ESS},n,t}^{\text{dc}} - p_{\text{ESS},n,t}^{\text{ch}}) - p_{j,t}^{\text{D}} \quad (92)$$

$$q_{j,t} = \sum_{g \in \Phi_{\text{G},j}} q_{g,t}^{\text{G}} + \sum_{n \in \Phi_{\text{RG},j}} q_{n,t}^{\text{RG}} + \sum_{n \in \Phi_{\text{ESS},j}} q_{\text{ESS},n,t}^{\text{SVC}} + \sum_{n \in \Phi_{\text{SVC},j}} q_{n,t}^{\text{SVC}} - q_{j,t}^{\text{D}} \quad (93)$$

where $p_{g,t}^{\text{G}}$ and $q_{g,t}^{\text{G}}$ are the active and reactive power output of the g^{th} generator at time t . $p_{n,t}^{\text{RG}}$ and $q_{n,t}^{\text{RG}}$ are the active and reactive power outputs of the n^{th} RG at time t . $q_{\text{ESS},n,t}^{\text{SVC}}$ and $q_{n,t}^{\text{SVC}}$ denote the reactive power outputs of the n^{th} ESS and the n^{th} SVC at time t , respectively. $p_{j,t}^{\text{D}}$ and $q_{j,t}^{\text{D}}$ are the active and reactive power demands of the load at bus j and time t . $\Phi_{\text{G},j}$, $\Phi_{\text{RG},j}$, $\Phi_{\text{ESS},j}$ and $\Phi_{\text{SVC},j}$ are the index set of generators, RGs, ESSs and SVCs connected to bus j .

2) Other operational constraints

The operational constraints of generators are presented as:

$$\underline{p}_g^G \leq p_{g,t}^G \leq \bar{p}_g^G, \forall g \in \Phi_G \quad (94)$$

$$\underline{Q}_g^G \leq q_{g,t}^G \leq \bar{Q}_g^G, \forall g \in \Phi_G \quad (95)$$

$$-RD_g \Delta t \leq p_{g,t+1}^G - p_{g,t}^G \leq RU_g \Delta t \quad (96)$$

$$0 \leq R_{g,t}^{G+} \leq RU_g \Delta t, \quad (97)$$

$$R_{g,t}^{G+} \leq \bar{p}_g^G - p_{g,t}^G, \forall g \in \Phi_G$$

$$0 \leq R_{g,t}^{G-} \leq RD_g \Delta t, \quad (98)$$

$$R_{g,t}^{G-} \leq p_{g,t}^G - \underline{p}_g^G, \forall g \in \Phi_G$$

$$\sum_{g \in \Phi_G} R_{g,t}^{G+} + \sum_{n \in \Phi_{ESS}} (R_{n,t}^{ch,ESS+} + R_{n,t}^{dc,ESS+}) \geq SRU_t, \quad (99)$$

$$\sum_{g \in \Phi_G} R_{g,t}^{G-} + \sum_{n \in \Phi_{ESS}} (R_{n,t}^{ch,ESS-} + R_{n,t}^{dc,ESS-}) \geq SRD_t$$

where $\underline{p}_g^G$ and $\bar{p}_g^G$ are the lower and upper bounds of the active power output of the g^{th} generator. $\underline{Q}_g^G$ and $\bar{Q}_g^G$ represent the lower and upper bounds of the reactive output of the g^{th} generator. RU_g and RD_g denote the upward and downward ramp rates of the g^{th} generator. SRU_t and SRD_t are the system-wide upward and downward reserve capacity requirements, which can be determined based on the historical uncertainty data of renewable energy and loads [5].

The operational constraints of RGs are given as follows:

$$\underline{p}_{n,t}^{RG} \leq p_{n,t}^{RG} \leq \bar{p}_{n,t}^{RG}, \forall n \in \Phi_{RG} \quad (100)$$

$$(p_{n,t}^{RG})^2 + (q_{n,t}^{RG})^2 \leq (S_n^{RG})^2, \forall n \in \Phi_{RG} \quad (101)$$

where $\bar{p}_{n,t}^{RG}$ is the predicted active power of the n^{th} RG at time t . $\underline{p}_{n,t}^{RG}$ is the lower bound of the active power output of the n^{th} RG at time t . And S_n^{RG} is the apparent power of the n^{th} RG.

The operational constraints of SVCs are described as:

$$\underline{Q}_n^{SVC} \leq q_{n,t}^{SVC} \leq \bar{Q}_n^{SVC}, \forall n \in \Phi_{SVC} \quad (102)$$

where $\underline{Q}_n^{SVC}$ and $\bar{Q}_n^{SVC}$ are the lower and upper bounds of the reactive power outputs of the n^{th} SVC, respectively.

The system operation also needs to meet the voltage security constraints and line transmission capacity limits.

$$\underline{V}_i \leq V_{i,t} \leq \bar{V}_i, \forall i \in \mathcal{N} \quad (103)$$

$$P_{ij,t}^2 + Q_{ij,t}^2 \leq \bar{S}_{ij}^2, \forall ij \in \mathbb{E} \quad (104)$$

where $\underline{V}_i$ and $\bar{V}_i$ are the lower and upper bounds of voltage magnitude of bus i , respectively. $\bar{S}_{ij}$ is the transmission capacity of branch ij .

Appendix E. Analyzing the applicability of the proposed relaxation condition for ESSs in the multi-period ED scenario

To elaborate on the applicability of the proposed relaxation condition across various scenarios, below, we will demonstrate the applicability of the proposed relaxation condition in the **multi-period ED scenario**.

Before providing the specific proof, we first give the mathematical model for multi-period ED based on the DC power flow model.

min (84).

s.t. (6)–(7), (10)–(16), (94), (96)–(100)

$$\sum_{j \in \mathcal{N}} \left(\sum_{g \in \Phi_{Gj}} p_{g,t}^G + \sum_{n \in \Phi_{RGj}} p_{n,t}^{RG} + \sum_{n \in \Phi_{ESSj}} (p_{n,t}^{dc,ESS} - p_{n,t}^{ch,ESS}) - p_{j,t}^D \right) = 0, \quad (\lambda_{j,t}^p) \quad (105)$$

$$-\bar{P}_{L,i} \leq \sum_{j \in \mathcal{N}} SF_{i-j} \left(\sum_{g \in \Phi_{Gj}} p_{g,t}^G + \sum_{n \in \Phi_{RGj}} p_{n,t}^{RG} + \sum_{n \in \Phi_{ESSj}} (p_{n,t}^{dc,ESS} - p_{n,t}^{ch,ESS}) - p_{j,t}^D \right) \leq \bar{P}_{L,i} \quad (\lambda_{L,i,t}^1, \lambda_{L,i,t}^2) \quad (106)$$

where SF_{i-j} denotes the shift factor for bus j on line i . $\bar{P}_{L,i}$ is the transmission capacity of line i . $\lambda_{L,i,t}^2$ and $\lambda_{L,i,t}^1$ represent the multipliers corresponding to the upper and lower limits of line capacity characterized by shift factors.

For multi-period ED, the *Condition C1* is rewritten as follows:

$$C1^{ED} : LMP_{j,t} > \lambda_{j,t}^{p,C1} \quad (107)$$

where $LMP_{j,t} = \lambda_{j,t}^p + \sum_j SF_{j-i}(\lambda_{L,j}^1(t) - \lambda_{L,j}^2(t))$ represents the location marginal price (LMP). The expression of $\lambda_{j,t}^{p,C1}$ is given as follows:

$$\lambda_{j,t}^{p,C1} = \left[\frac{\partial obj}{\partial(p_{ESS,n,t}^{ch})} \middle/ \eta_{ESS,n}^{dc} + \frac{\partial obj}{\partial(p_{ESS,n,t}^{dc})} \cdot \eta_{ESS,n}^{ch} \right] \middle/ \left(\eta_{ESS,n}^{ch} - \frac{1}{\eta_{ESS,n}^{dc}} \right) \quad (108)$$

Proof of Condition C1^{ED}: Denoting the Lagrangian function of the relaxed ED model as L_{RM}^{ED} , and defining a new variable $\Gamma_{ESS,n,t} = \sum_{t=t}^T (1 - \delta_{ESS,n})^{t-t} (\lambda_{ESS,n,t}^{SOC,2} - \lambda_{ESS,n,t}^{SOC,1})$. Employing KKT conditions, the following equations are obtained:

$$\frac{\partial L_{RM}}{\partial(p_{ESS,n,t}^{ch})} = \frac{\partial obj}{\partial(p_{ESS,n,t}^{ch})} + LMP_{j,t} - \lambda_{ESS,n,t}^{ch,1} + \lambda_{ESS,n,t}^{ch,2} + \frac{\lambda_{ESS,n,t}^{relax}}{\bar{P}_{ESS,n}^{ch}} + \eta_{ESS,n}^{ch} \cdot \Gamma_{ESS,n,t} \cdot \Delta t + \lambda_{ESS,n,t}^{RD,1} = 0 \quad (109)$$

$$\frac{\partial L_{RM}}{\partial(p_{ESS,n,t}^{dc})} = \frac{\partial obj}{\partial(p_{ESS,n,t}^{dc})} - LMP_{j,t} - \lambda_{ESS,n,t}^{dc,1} + \lambda_{ESS,n,t}^{dc,2} + \frac{\lambda_{ESS,n,t}^{relax}}{\bar{P}_{ESS,n}^{dc}} - \Gamma_{ESS,n,t} \cdot \Delta t \middle/ \eta_{ESS,n}^{dc} + \lambda_{ESS,n,t}^{RU,1} = 0 \quad (110)$$

With (109)/ $\eta_{ESS,n}^{dc}$ + (110)· $\eta_{ESS,n}^{ch}$, we have

$$\left[\frac{\partial obj}{\partial(p_{ESS,n,t}^{ch})} + LMP_{j,t} - \lambda_{ESS,n,t}^{ch,1} + \lambda_{ESS,n,t}^{ch,2} + \frac{\lambda_{ESS,n,t}^{relax}}{\bar{P}_{ESS,n}^{ch}} + \lambda_{ESS,n,t}^{RD,1} \right] \middle/ \eta_{ESS,n}^{dc} + \left[\frac{\partial obj}{\partial(p_{ESS,n,t}^{dc})} - LMP_{j,t} - \lambda_{ESS,n,t}^{dc,1} + \lambda_{ESS,n,t}^{dc,2} + \frac{\lambda_{ESS,n,t}^{relax}}{\bar{P}_{ESS,n}^{dc}} + \lambda_{ESS,n,t}^{RU,1} \right] \cdot \eta_{ESS,n}^{ch} = 0 \quad (111)$$

Further, eq. (111) yields:

$$\begin{aligned} LMP_{j,t} = & \left[\frac{\partial obj}{\partial(p_{ESS,n,t}^{ch})} \middle/ \eta_{ESS,n}^{dc} + \frac{\partial obj}{\partial(p_{ESS,n,t}^{dc})} \cdot \eta_{ESS,n}^{ch} \right] \middle/ \left(\eta_{ESS,n}^{ch} - \frac{1}{\eta_{ESS,n}^{dc}} \right) \\ & - \left\{ \left[\lambda_{ESS,n,t}^{ch,2} + \frac{\lambda_{ESS,n,t}^{relax}}{\bar{P}_{ESS,n}^{ch}} + \lambda_{ESS,n,t}^{RD,1} \right] \middle/ \eta_{ESS,n}^{dc} \right. \\ & + \left. \left[\lambda_{ESS,n,t}^{dc,2} + \frac{\lambda_{ESS,n,t}^{relax}}{\bar{P}_{ESS,n}^{dc}} + \lambda_{ESS,n,t}^{RU,1} \right] \cdot \eta_{ESS,n}^{ch} \right\} \middle/ \left(\frac{1}{\eta_{ESS,n}^{dc}} - \eta_{ESS,n}^{ch} \right) \\ & + \left(\lambda_{ESS,n,t}^{ch,1} \middle/ \eta_{ESS,n}^{dc} + \lambda_{ESS,n,t}^{dc,1} \cdot \eta_{ESS,n}^{ch} \right) \middle/ \left(1 \middle/ \eta_{ESS,n}^{dc} - \eta_{ESS,n}^{ch} \right) \end{aligned} \quad (112)$$

In combination with the expression of $LMP_{j,t}$ in eq. (112), we can conclude that if *Condition C1^{ED}* holds, we can obtain (113).

$$\begin{aligned} & \left(\lambda_{ESS,n,t}^{ch,1} \middle/ \eta_{ESS,n}^{dc} + \lambda_{ESS,n,t}^{dc,1} \cdot \eta_{ESS,n}^{ch} \right) \middle/ \left(1 \middle/ \eta_{ESS,n}^{dc} - \eta_{ESS,n}^{ch} \right) > \\ & \left\{ \left[\lambda_{ESS,n,t}^{ch,2} + \frac{\lambda_{ESS,n,t}^{relax}}{\bar{P}_{ESS,n}^{ch}} + \lambda_{ESS,n,t}^{RD,1} \right] \middle/ \eta_{ESS,n}^{dc} + \left[\lambda_{ESS,n,t}^{dc,2} + \frac{\lambda_{ESS,n,t}^{relax}}{\bar{P}_{ESS,n}^{dc}} + \lambda_{ESS,n,t}^{RU,1} \right] \cdot \eta_{ESS,n}^{ch} \right\} \middle/ \left(\frac{1}{\eta_{ESS,n}^{dc}} - \eta_{ESS,n}^{ch} \right) \end{aligned} \quad (113)$$

Considering $\eta_{ESS,n}^{dc} \cdot \eta_{ESS,n}^{ch} > 0$, $\lambda_{ESS,n,t}^{ch,2}, \lambda_{ESS,n,t}^{dc,2} \geq 0$ and $\lambda_{ESS,n,t}^{relax}, \lambda_{ESS,n,t}^{RD,1}, \lambda_{ESS,n,t}^{RU,1}, p_{ESS,n,t}^{ch}, p_{ESS,n,t}^{dc} \geq 0$, we can derive

$$\left[\lambda_{ESS,n,t}^{ch,2} + \frac{\lambda_{ESS,n,t}^{relax}}{\bar{P}_{ESS,n}^{ch}} + \lambda_{ESS,n,t}^{RD,1} \right] \middle/ \eta_{ESS,n}^{dc} + \left[\lambda_{ESS,n,t}^{dc,2} + \frac{\lambda_{ESS,n,t}^{relax}}{\bar{P}_{ESS,n}^{dc}} + \lambda_{ESS,n,t}^{RU,1} \right] \cdot \eta_{ESS,n}^{ch} \geq 0 \quad (114)$$

Combining formulas (113), (114) and $1/\eta_{ESS,n}^{dc} > \eta_{ESS,n}^{ch}$, we have

$$\left(\lambda_{ESS,n,t}^{ch,1} \middle/ \eta_{ESS,n}^{dc} + \lambda_{ESS,n,t}^{dc,1} \cdot \eta_{ESS,n}^{ch} \right) > 0 \quad (115)$$

Since $\eta_{ESS,n}^{dc} \cdot \eta_{ESS,n}^{ch} > 0$ and $\lambda_{ESS,n,t}^{ch,1}, \lambda_{ESS,n,t}^{dc,1} \geq 0$, (116) holds.

$$\lambda_{ESS,n,t}^{ch,1} + \lambda_{ESS,n,t}^{dc,1} > 0 \quad (116)$$

Formula (116) indicates that $p_{ESS,n,t}^{ch} \cdot p_{ESS,n,t}^{dc} = 0$. In other words, based on *Condition C1^{ED}*, the optimal solution enforces the complementary constraint, even though it is relaxed.

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